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Interactive Mapping Application for Environmental Monitoring and Forecasting

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Abstract

This research paper provides an environmental monitoring and forecasting model based on publicly available DEFRA UK-AIR data for the study and prediction of NO₂ concentrations in the UK. Applying a quantitative methodology, this project covers data preparation, data exploration, temporal and historical feature engineering, regression modelling, explainable artificial intelligence, spatial clustering and building of a web application. The four models-Linear Regression, Decision Tree, Random Forest, and XGBoost-have been built and evaluated through their MAE, RMSE and coefficient of determination (R²). A chronological train-test split has been used to ensure temporal ordering.

The first attempts to build the model brought unexpectedly good results, including R² of 1.0000 for the Linear Regression model. Validation has shown that there was a problem of target leakage in the original rolling NO₂ feature because it included the target observation itself. Therefore, historical features have been engineered and the rolling mean feature has been excluded. In the configuration of the final model with six features, the Random Forest model demonstrated the best performance with MAE 3.8748, RMSE 5.9533 and R² 0.8529. The final version of Random Forest model has been chosen for SHAP-based explainability and the Flask application development.

SHAP analysis was used to evaluate the influence of the features on NO₂ predictions, while the K-Means clustering technique was used to discover spatial patterns and pollution hotspots relatively. Five clusters were created, out of which Cluster 3 stands out as the hotspot since it returned the highest average value of NO₂ concentration of about 18.71 µg/m³. Folium made possible the visual representation of monitoring stations and clustering results interactively. The resulting Flask prototype combines NO₂ prediction, explanation of model predictions, and spatial environment analysis into a single interface.

To conclude, the paper provides an example of how machine learning, explainable AI, and spatial analysis can be combined in order to build an environmental monitoring system. The importance of correcting for target leakage during feature engineering was highlighted. The presented system is still a local prototype, and more thorough validation should be done before its deployment.

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List of Abbreviations

Abbreviation	Full Form
AI	Artificial Intelligence
API	Application Programming Interface
CSV	Comma-Separated Values
DBSCAN	Density-Based Spatial Clustering of Applications with Noise
DEFRA	Department for Environment, Food and Rural Affairs
EDA	Exploratory Data Analysis
HTML	HyperText Markup Language
K-Means	K-Means Clustering
MAE	Mean Absolute Error
ML	Machine Learning
NO₂	Nitrogen Dioxide
O₃	Ozone
PM₁₀	Particulate Matter with a Diameter of 10 Micrometres or Less
PM_{2.5}	Particulate Matter with a Diameter of 2.5 Micrometres or Less
R²	Coefficient of Determination
RF	Random Forest
RMSE	Root Mean Square Error
SHAP	SHapley Additive exPlanations
SO₂	Sulphur Dioxide
UK	United Kingdom
UK-AIR	UK Air Information Resource
XAI	Explainable Artificial Intelligence
XGBoost	Extreme Gradient Boosting

Chapter 1: Introduction

1.1 Background

Air pollution is a significant environmental and public health issue, with the pollutants including nitrogen dioxide (NO_2) that fluctuate in different areas and at different times. It is important to track these changes because they help to assess pollutant patterns and aid in environmental management. The project utilises publicly available data provided by the DEFRA UK-AIR monitoring network, which includes pollutant data from monitoring stations throughout the United Kingdom. With the advent of large environmental datasets, there is an opportunity for data-driven analysis, but the datasets may be hard for non-technical users to interpret. The project proposal recognises that environmental monitoring data is currently available in technical formats and that many environmental monitoring systems exist that provide only past measurements, but not predictive information.

Machine learning can be used for the purposes of environmental forecasting by being able to find correlations between certain historical observations and the concentration of pollutants. In this project, several regression models are built and tested to estimate the amount of NO_2 .

1.2 Research Problem

The primary research challenge is the disconnect between environmental data and predictive modelling, spatial analysis and application to the user. The literature shows that the ability to forecast, detect hotspots, explainability and interactive mapping are not typically integrated in a single Operational System, built with official environmental monitoring data. This makes the added value of predictive environmental intelligence for users who don't have the technical expertise in machine learning or raw monitoring data rather difficult to access.

1.3 Research Aim

The aim of this research is to develop and evaluate a spatio-temporal environmental monitoring and forecasting system using UK air-quality data, machine learning techniques, spatial hotspot analysis and an interactive web-based application.

1.4 Research Objectives

- Analyze spatial and temporal patterns in air pollution using DEFRA UK-AIR monitoring data.
- Develop and evaluate regression models, including Linear Regression, Decision Tree, Random Forest and XGBoost, to predict NO_2 concentrations.

- Identify relative pollution hotspots using spatial clustering of monitoring locations.
- Apply explainable AI techniques to interpret the contribution of features to machine-learning predictions.
- Develop a Flask-based web application integrating NO₂ prediction, explainability and spatial analysis.

1.5 Research Questions

This research investigates the effectiveness of machine learning models in predicting NO₂ concentrations using temporal and historical air-quality features from DEFRA UK-AIR data. It examines which machine learning approach provides the most suitable predictive performance for the selected NO₂ prediction task and explores how SHAP can improve the interpretability of machine learning predictions for environmental monitoring. The study also investigates how spatial clustering can be used to identify relative geographical hotspots of NO₂ concentration across monitoring locations. Finally, it examines how prediction, explainability and spatial analysis can be integrated into a functional Flask-based environmental monitoring application.

1.6 Scope of the Research

The project includes the analysis of UK air-quality data, creation of temporal and historical features, development and evaluation of regression models, interpretation of model behaviour using SHAP, spatial analysis using K-Means clustering, and integration of the analytical results into a Flask application. The application provides estimated NO₂ concentrations using the selected Random Forest model and presents supporting model and geographical information. The system was developed as a local prototype rather than a production-ready application. The project does not attempt to establish causal relationships between individual features and NO₂ concentrations.

1.7 Significance of the Research

The significance of this study lies in bringing environmental data analysis and an interactive application together within a single project. Rather than treating prediction, explainability and spatial analysis as separate tasks, the study combines these components within an integrated framework. The Flask application provides a practical interface for generating NO₂ predictions and accessing supporting model and geographical information. This demonstrates how machine learning and spatial analysis can be brought together to make environmental monitoring results more accessible and easier to interpret.

Chapter 2: Literature Review

2.1 Introduction

Air pollution remains a major environmental and public health concern, with pollutants contributing to respiratory and cardiovascular illnesses as well as wider environmental impacts. Although national environmental monitoring systems have increased the availability of air-quality data, greater data availability does not necessarily translate into greater public understanding or effective decision-making (Manal Karmoude et al., 2025). Existing approaches often concentrate on data distribution and visualisation rather than transforming monitoring data into predictive and interpretable environmental information (Mujtaba et al., 2025; Rahman et al., 2024). The recent advances in machine learning (ML) have depicted tremendous strides in forecasting air pollutant concentrations through learning complex and non-linear relationships in environmental data (Alrashidi et al., 2025; Natarajan et al., 2024). Simultaneously, time and space analyses of the trends of pollution have shown that there is a significant degree of geographic variation, which makes spatial intelligence significant in environmental assessment (Morawiec et al., 2026). Also, the development of explainable artificial intelligence (XAI) has helped address the growing issues of model transparency and confidence in decision-making (Ma et al., 2023).

There are statistical models that provide ease of use and interpretation, and on the other hand, there are machine learning or ensemble models that can account for non-linear associations between the variables at the expense of increased computation and lower model interpretability.

2.2 Air Quality Monitoring and Open Environmental Data

The presence of air quality monitoring facilities has significantly increased to high-resolution environmental data, especially in national and city monitoring systems that measure major pollutants like PM 2.5, PM 10, NO₂, and O₃. These systems are important to assist in environmental governance, compliance with regulations, and educating people about health strategies. Such datasets can be further enhanced by open-access initiatives that will lead to a higher level of transparency and ease the analysis of various data (Manal Karmoude et al., 2025). But the assumption that greater data availability automatically results in better environmental knowledge is also open to challenge (Lilhore et al., 2025)

Air quality datasets, even when large and abundant, are typically described by high structural complexity, such as heterogeneous data formats, lack of data, and high temporal granularity, which reduce their usefulness outside of communities of experts. Mujtaba et al. (2025) are of the opinion that although such datasets are useful in predictive modelling, their practical application is limited because of the unavailability of analytical tools that can be used to convert raw measurements into usable information. As a result, current monitoring systems are more likely to focus on data

dissemination than on meaningful interpretation (Mujtaba et al., 2025; Rahman et al., 2024).

There are certain demands put forward by the use of predictive systems, including validation in time, feature extraction, and interpretation of the model. Hence, an adequate environmental monitoring system should combine availability of data and predictions with clear analytics.

2.3 Machine Learning for Air Quality Forecasting

2.3.1 Traditional and Baseline Approaches

Statistical and time-series models were used as predecessors of early air quality forecasting because they assume that there are linear relationships between variables and that the time series is stable. Although these are interpretable and simple in principle, limited in capturing the complex, non-linear interactions between the meteorological factors, emissions and seasonal variations that define air pollution dynamics (Al Yammahi & Aung, 2023).

Greater flexibility does not automatically imply better forecasting performance across all datasets. Differences in dataset characteristics, forecasting horizons and feature availability can substantially influence model performance. Therefore, statistical models should not be considered obsolete, but rather provide a useful benchmark against which more complex machine learning approaches can be evaluated.

2.3.2 Data Preparation and Feature Engineering

The data in air quality is prone to contain missing values, inaccurate measurements from sensors, outliers, and inconsistent time periods, making data preprocessing an integral part of the modelling process. According to Natarajan et al. (2024), preprocessing methods, including missing value imputation, normalisation of the data, and feature selection, have significantly increased the accuracy of the prediction by reducing noise. Similarly, Manal Karmoude et al. (2025) note data cleansing and transformations as necessary stages for accurate machine learning predictions. Feature engineering has also been shown to be an important factor in forecast precision (Liu et al., 2024). In the existing literature, the use of time lags, moving average values, rolling statistics, seasonality indicators, and weather-related parameters is common in better predicting pollutants' behaviour (Natarajan et al., 2024; Mujtaba et al., 2025). Feature engineering allows for modelling complex temporal dynamics that raw measures cannot reveal by themselves.

Though recognising the significance of preprocessing and feature generation, many studies provide inadequate attention to the methods used for generating the features, as well as the decision criteria used for choosing the features (Kalaiselvi et al., 2025).

The use of temporal characteristics gives rise to an important methodological issue. While lagged variables and rolling statistics improve the predictive capacity of a model

using information from earlier time points, it is essential that they be built on the basis of a preserved temporal ordering of the data.

2.3.3 Machine Learning Models

There has been increasing use of machine learning models to predict air quality as these models are capable of identifying nonlinear relationships between pollutant concentrations, meteorological conditions, and other environmental factors. But the performance is dependent on various factors including the nature of the data set used, selected features, time period of forecasting and the structure of the machine learning algorithm itself. Thus, it becomes more enlightening to compare different machine learning algorithms in terms of their performance rather than assuming that a particular algorithm always outperforms others (Koziel et al., 2024; Alrashidi et al., 2025; Natarajan et al., 2024).

Various comparative studies reveal that gradient boosting methods provide an edge over other simpler machine learning algorithms in case of very nonlinear relationships between various environmental variables (Tırınk, 2025; Varshney et al., 2025). But there should be trade-offs between better performance and increased complexity. For example, Makhdoomi et al. (2025) have shown the potential of more complex models for the purpose of virtual monitoring stations and that predictive performance can lead to wider environmental applications.

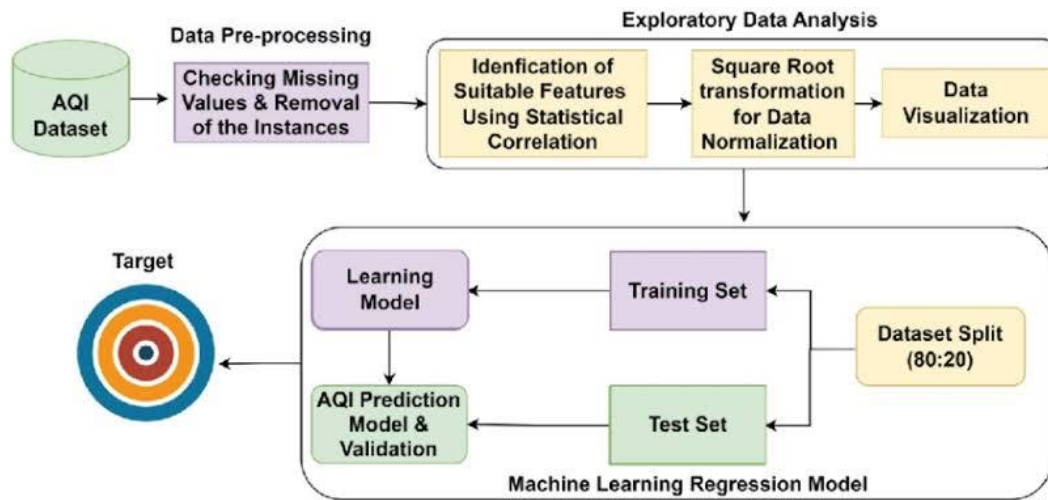


Figure 1: Machine Learning Workflow for Air Quality Prediction

(Source: Natarajan et al., 2024)

The figure shows the entire pipeline in end-to-end AQI prediction, such as data processing (preprocessing), feature selection, model training, and validation.

Despite these achievements, machine learning models are most sensitive to a dataset's properties, such as regional variability and data quality. This brings up the issue of their generalisability in varying geographical settings. In addition, albeit with good predictive performance, the models are commonly only applied in experimental studies, and there is little translation into operational environmental monitoring systems.

2.3.4 Ensemble and Advanced Learning Models

To continue improving predictive performance, recent research has examined ensemble and hybrid learning systems, such as boosting, stacking, and mixed systems. The same applies to Xu et al. (2025), who put a good emphasis on the efficacy of stacking models to capture a variety of data patterns, leading to high-performance of stacking models as opposed to single-model designs.

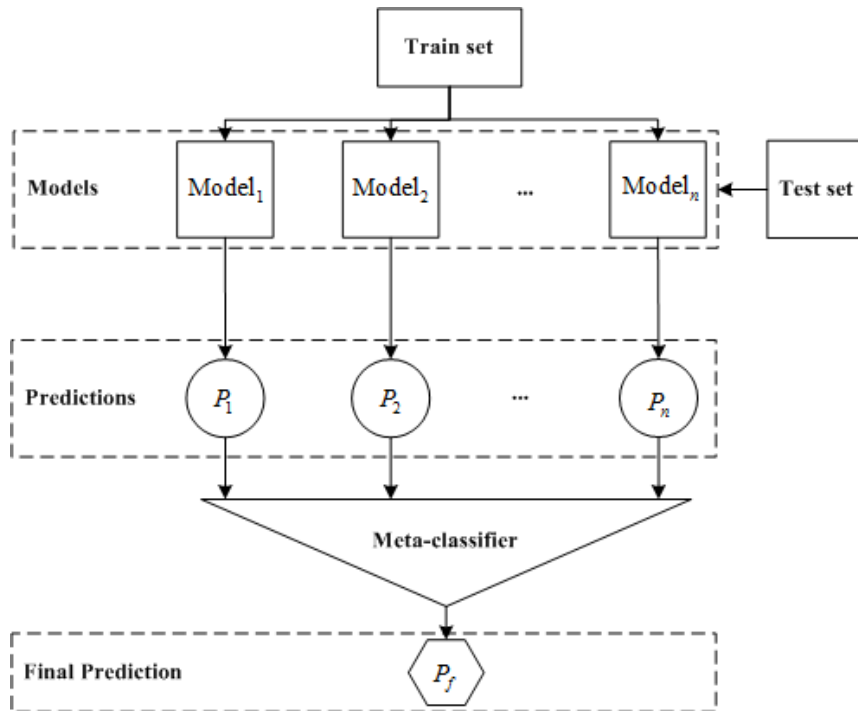


Figure 2: Stacking Ensemble Learning Architecture

(Source: Singh et al., 2026)

The figure demonstrates how several base models can be combined using a meta-classifier to enhance predictive performance.

Ensemble learning entails a compromise between prediction ability, computational resources, and interpretability of the machine learning models. While one single machine learning model could be simpler to develop and interpret, ensemble approaches such as stacking include a combination of several learners in order to model various trends in the data. Although it leads to better predictions in some cases, the increased complexity makes it difficult to understand the way that the model operates (Yenkikar et al., 2025).

2.3.5 Limitations of Current Machine Learning Research

Most works focus on performance measures and neglect interpretability, integration, and user accessibility (Singh et al., 2026; Xu et al., 2025). This piecemeal methodology culminates in standalone innovations that cannot be implemented into holistic environmental intelligence solutions (Havaei et al., 2025). As such, although machine learning methodology has made great progress over time, its impact on practice remains limited, underscoring the need for integrated, explainable, and user-friendly solutions.

The other limitation concerns the ability to compare and verify the predictive accuracy. A very high level of accuracy does not guarantee good generalization if the process of evaluation does not ensure the appropriate maintenance of temporal independence or if the predictors contain information which is not available at the moment of prediction.

2.4 Spatio-Temporal Modelling and Hotspot Detection

2.4.1 Spatial Variability of Air Pollution

The spatial heterogeneity of air pollution is high due to variations in urban morphology, traffic levels, industrial presence, and meteorological conditions (Morawiec et al., 2026; Dar & Shaik, 2025). Rather than even distributions, the concentration of pollutants is generally concentrated at local scales, especially within a specific area of geographic space, so that there are local sources of emission and environmental structures. According to Morawiec et al. (2026), if the spatial analysis is performed at a large scale, it has the opportunity to differentiate between the macro-regions of pollution, and this aspect demonstrates the importance of the geographic context in the course of environmental assessment.

However, a considerable portion of the present literature views spatial variability as an ancillary analytical dimension of predictive modelling rather than a key component (Kuhn et al., 2024). It diminishes the ability to capture local risk patterns, particularly in urban areas where pollution gradients may vary by many meters over a few meters (Kamigauti et al., 2024).

2.4.2 Clustering Techniques and Hotspot Detection

Ataee et al. (2026) present a correlation-distance index to enhance clustering accuracy in sensor networks, demonstrating that advanced spatial methods can improve data reliability. Similarly, Dar and Shaik (2025) combine spatio-temporal clustering and forecast models, and disclose that hotspot detection can be of great benefit to understand the areas with high risks and the dynamics of pollution.

There is always a need for methodology considerations when choosing clustering methods. K-means is easy to implement and efficient but requires setting the number of clusters beforehand and performs poorly with clusters having non-standard shapes. On the other hand, DBSCAN is able to recognize irregular clusters and even find outliers without requiring the pre-defined number of clusters; nonetheless, it depends on parameter settings and data density.

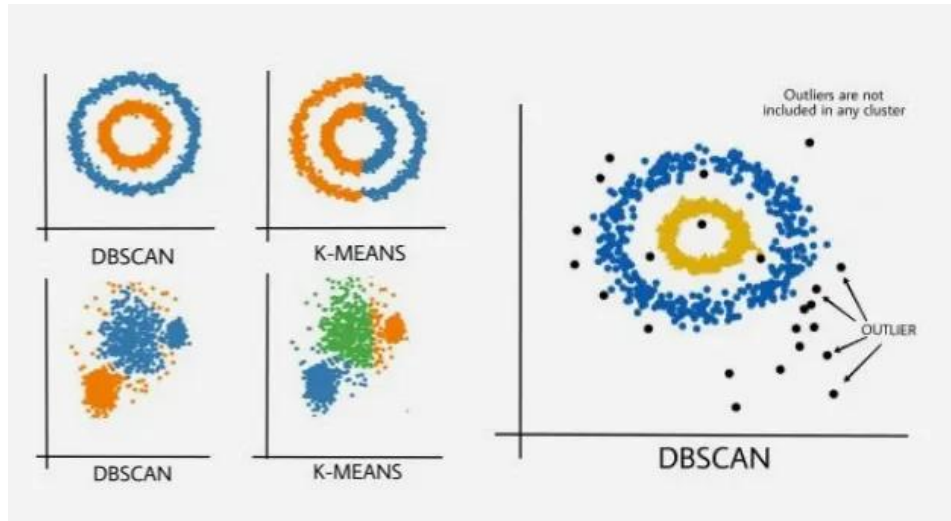


Figure 3: Clustering-Based Air Pollution Hotspot Detection

(Source: Dar & Shaik, 2025)

In spite of these contributions, the clustering methods are frequently applied in isolation, with the main emphasis on pattern detection, instead of their integration with predictive modelling approaches (Choi et al., 2024). This leads to the stagnation of hotspot descriptions that do not account for temporal dynamics and cannot be useful for guiding future environmental management (Yasin et al., 2025).

2.4.3 Integration Challenges in Spatio-Temporal Modelling

One of the most important weaknesses of the literature is the continuing divide between spatial analysis and temporal forecasting (Kho et al., 2026). Subsequently, the available literature provides piecemeal information, further underscoring the importance of combining spatio-temporal models to fill this void and enable more comprehensive decisions about the environment.

2.5 Explainable AI in Air Quality Modelling

2.5.1 Need for Explainability in Environmental Models

The growing use of machine learning to predict air quality has heightened concerns about the transparency of predictive models. Most successful algorithms are so-called black boxes, where they are accurate but do not provide a person with insight into the algorithm's decision-making (Houdou et al., 2024). This absence of transparency is especially problematic in the environmental context, because it is often the predictions that inform the interventions of population health and policy-making decisions. According to Ma et al. (2023), interpretable models may be used to show the impact of external variables, such as policy interventions, on the level of pollution and increase the credibility of the analytical results. Muhammad et al. (2025) indicate that the stakeholders who are interested in the model prediction need the model predictions to be credible in addition to being interpretable, and are willing to make decisions based on them.

It poses a conflict between accuracy and trust, in which highly accurate models can still be inappropriate in practice decision-making, as their results cannot be explicated. Therefore, explainability does not appear only as a technical improvement, but it is an essential condition of the responsible implementation of AI in environmental systems.

2.5.2 SHAP and Interpretability Techniques

To overcome these issues, explainable artificial intelligence methods such as Shapley Additive explanations (SHAP) have become popular. SHAP provides global and local interpretability by measuring the impact of individual features on model predictions. The research by Singh et al. (2026) and Xu et al. (2025) shows that SHAP has the capacity to reveal complicated interactions between environmental factors and pollution levels, which allow a more sophisticated interpretation of model behaviour.

Even with its strength in analysis, the SHAP application is usually limited to post hoc interpretation in experimental studies (Birinci et al., 2026). Despite being one of the most popular explainability methods today, SHAP certainly comes with its share of shortcomings (Jiménez-Navarro et al., 2024). Firstly, the computing expense associated with the method increases significantly depending on the complexity of the model being used. Moreover, SHAP-generated explanations could also be hard to understand without the relevant background knowledge.

Methods of feature importance provide simpler global information about the important variables while SHAP offers both local and global information for the particular prediction. SHAP involves more computation and its results can be difficult for laypersons to understand (Jiménez-Navarro et al., 2024). Therefore, when choosing the explainability technique, there needs to be consideration of the complexity of the result and ease of understanding the same.

2.5.3 Limitations of Current XAI Applications

One of the main weaknesses of the existing research is that explainability is discussed more as an analytical step than as an essential part of operating systems (He et al., 2024). XAI methods are not often implemented in interactive platforms that can enable users to explore and analyse predictions in real-time (Singh et al., 2026; Xu et al., 2025).

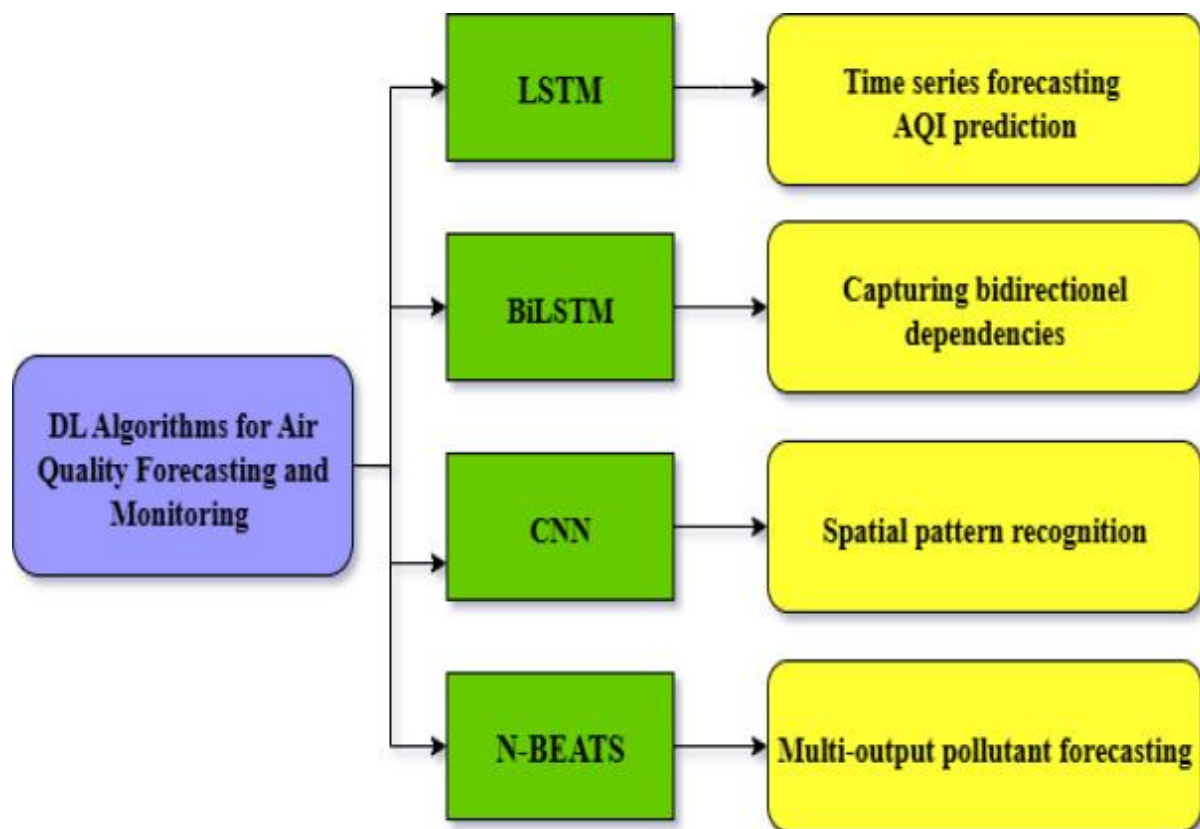


Figure 4: Most commonly used and high-performing deep learning algorithms

(Source: Manal Karmoude et al., 2025)

2.6 Environmental Monitoring Systems and Interactive Platforms

Such systems are usually used to pool real-time and historical information and display pollutant levels using visual interfaces so as to increase awareness and transparency. Rahman et al. (2024) show how machine learning models can be made available on web-based platforms that allow users to interact with predictive data. On the same note, the models presented by Rajesh et al. (2025) are for real-time monitoring and predictive mapping; they are applicable in the analysis of environmental health.

Even with these developments, the majority of the current platforms are simply limited in scope and functionality (Jalali et al., 2025). The main limitation observed in existing systems for monitoring is the descriptive nature of such systems, focusing more on historical visualisation than on predictive decision support (Rahman et al., 2024; Rajesh et al., 2025). In many cases, even when they are included in the interface, predictive models are not integrated fully into the user interface and remain inaccessible and unusable. Moreover, there is a low degree of explainability so that users cannot comprehend the drivers of the predicted level of pollution (Rahman et al., 2024; Rajesh et al., 2025).

The foundational dashboard provides relatively simple information with little computational power needed, while the system designed for predictive purposes requires additional modeling and validation components. Even though predictive tools could provide more value in decision-making, increased technical difficulties associated with the usage of these systems create issues concerning reliability and usability of such systems.

2.7 Summary Table of Literature

The following table summarises key studies, their methodological focus, contributions, and limitations:

Table 1: Summary Table

Study	Technique	Main Strength	Main Limitation	Critical Implication
Alrashidi et al. (2025)	ML models	Accurate AQI prediction	No spatial integration	Predictive accuracy alone does not provide geographically contextualised environmental intelligence.

Natarajan et al. (2024)	ML optimisation	Improved prediction accuracy	Limited generalisability	Performance may depend strongly on dataset and environmental context.
Tirink (2025)	Comparative ML models	Model benchmarking	No system integration	Comparative modelling is valuable, but model performance needs to be connected to practical system requirements.
Singh et al. (2026)	Ensemble + SHAP	Prediction and interpretability	Increased complexity	Higher predictive capability may introduce computational and interpretability trade-offs.
Xu et al. (2025)	Stacking + SHAP	Enhanced model insights	Computational cost	Ensemble-based explainability can improve analysis but may reduce efficiency.
Ataee et al. (2026)	Spatial clustering	Sensor/spatial pattern analysis	No forecasting capability	Spatial analysis alone cannot provide future pollution prediction.
Dar & Shaik (2025)	Spatio-temporal clustering	Hotspot identification	Limited deployment	Combining temporal and spatial information improves analytical coverage, but operationalisation remains challenging.
Rahman et al. (2024)	Web-based ML system	Interactive ML interface	Limited explainability	System accessibility does not necessarily

				ensure model transparency.
Rajesh et al. (2025)	Predictive mapping	Real-time environmental assessment	Weak user interpretability	Predictive functionality needs to be accompanied by understandable explanations.

The comparison reveals that the techniques discussed provide complimentary capabilities instead of competing with each other as substitutes. Both statistical and conventional machine learning models differ in their compromises between simplicity and flexibility in prediction, while ensembles can improve predictive power with increased computational requirements. In the same way, clustering algorithms provide spatial information that is missed by conventional forecasting models, while forecasting models provide time-related predictions which cannot be made using clustering only. SHAP helps to interpret results but increases computational effort and does not guarantee user-friendliness.

2.8 Conceptual Framework

The conceptual framework introduces an integrated architecture of a combination of data preprocessing, machine learning-based air quality prediction, spatial hotspot identification, and explainable AI in a unified system. It integrates SHAP to help improve model transparency and interpretability so that users can gain insight into the key influencing factors. The below framework is therefore designed to address the comparative limitations present in the literature by combining complementary analytical approaches rather than relying on prediction from a single model.

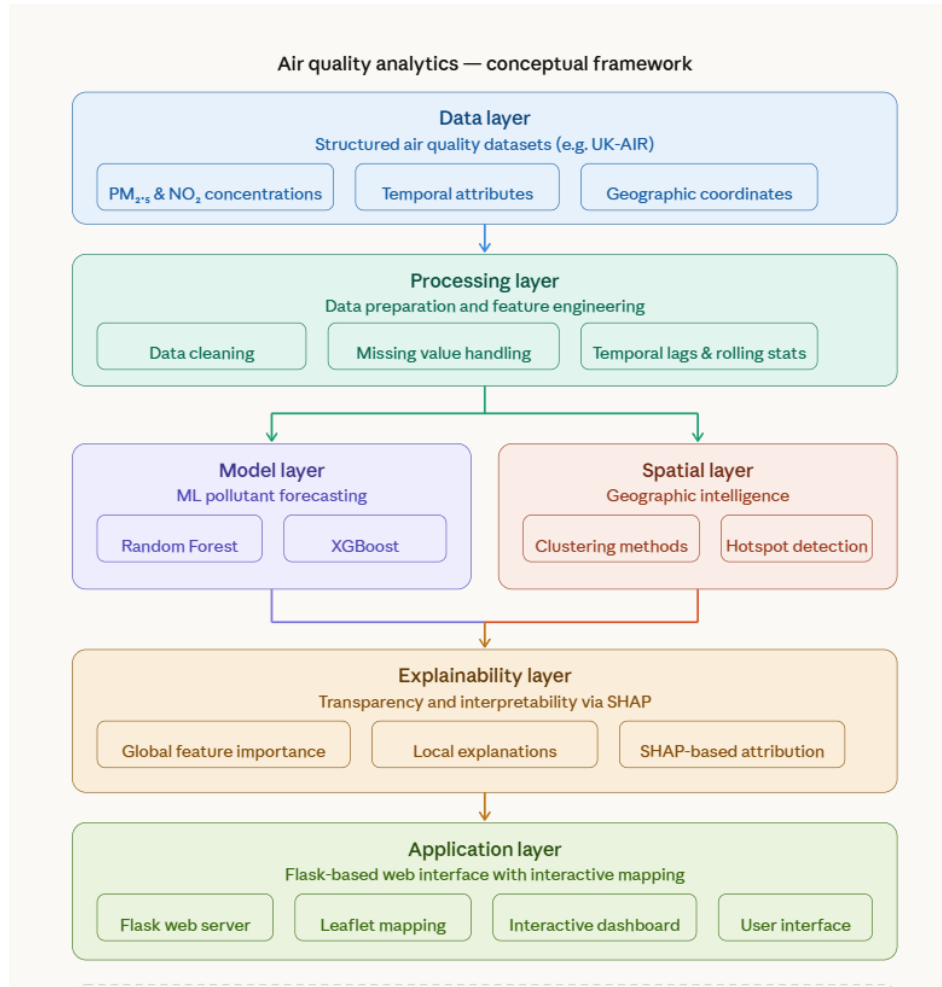


Figure 5: Conceptual Framework of the Proposed Air Quality Monitoring System
(Source: Self-Made)

2.9 Research Gap and Contribution

The present research is essentially fragmented, even though air quality forecasting, spatial, and explainable artificial intelligence have been improved significantly. The predictive performance of machine learning models is high, yet they are not integrated into systems and platforms used by users (Alrashidi et al., 2025; Tırink, 2025). Equally, spatial analysis tools are effective for locating pollution hotspots. However, they are mostly considered as analytical operations that are not integrated with models of time forecasting (Dar & Shaik, 2025).

The literature review highlights five major gaps in existing knowledge. First, predictive machine learning algorithms are rarely used together with spatial hot spot analysis. Second, prediction models are evaluated based on their predictive power, whereas other aspects, such as accuracy, model complexity, computation cost, and interpretability, are considered less often. Third, the SHAP method is used mostly for post hoc research purposes and not for operational needs. Fourth, environmental monitoring systems are geared towards visualization and information distribution and

do not focus enough on making decisions based on verified predictions. Fifth, little attention is paid to comprehensive systems combining forecasting, spatial analysis, explainability, user-friendly interfaces, and reliable temporal validation.

Chapter 3: Research Design & Methodology

3.1 Research Approach

The method used in this research is quantitative, data-driven, and applied research. The study can analyse the patterns of concentration of NO₂ in the air due to environmental monitoring data obtained from secondary sources and then produce a predictive model, and also determine the differences in air quality across geographic locations. The analytical process integrates data preprocessing, exploratory analysis, feature engineering, machine learning, explainable artificial intelligence and spatial clustering. The approach is also adopted because the research is not limited to assessing the machine learning model. An interactive interface is created using Flask to allow for access to the analytical results, with the selected model being incorporated in the Flask application. Therefore, the research links computational analysis and a practical prototype of environmental monitoring.

3.2 Research Design

The research model is an experimental and system development model. The experimental part entails the design of four regression algorithms to predict NO₂ and comparing their results. The system-development part brings together the resulting model, the SHAP explainability, the K-Means spatial analysis, the Folium visualisation and a Flask application. The modelling stage does not imply that a specific algorithm will give the best results. Linear Regression, Decision Tree, Random Forest and XGBoost are compared with the common regression metrics. This provides a basis for choosing which model is used in the prediction interface, based on the evidence. The methodology documentation highlights the importance of model comparison for the selection of the most appropriate way to forecast NO₂.

3.3 Proposed System Architecture

The overall architecture links the environmental datasets, preprocessing, machine learning, explainability, spatial analysis and web application.

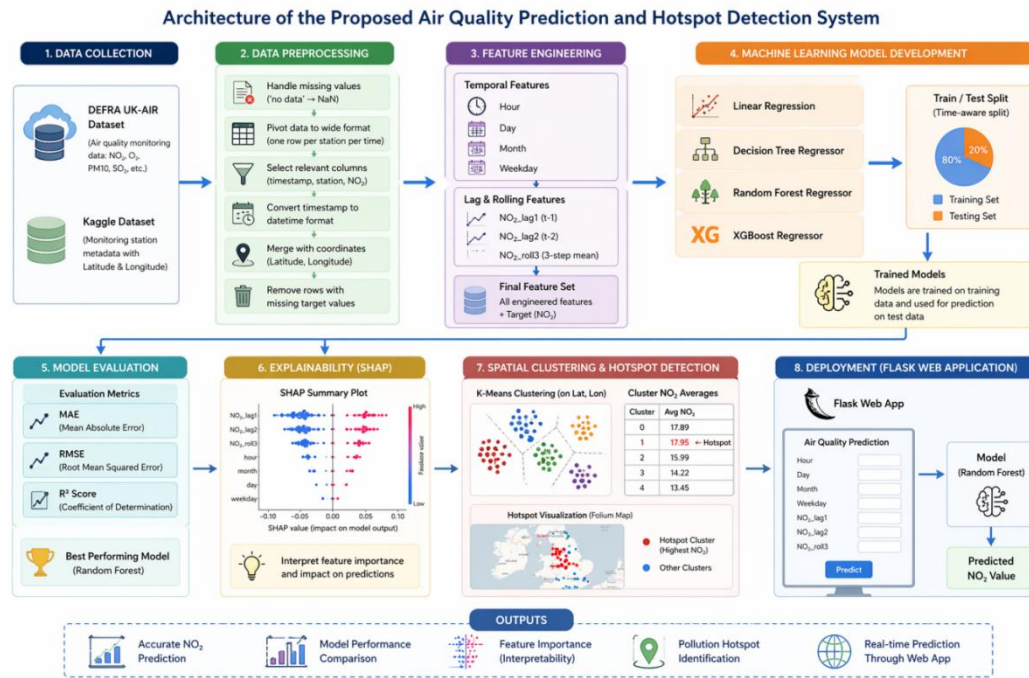


Figure 3.1: Architecture of the proposed air quality prediction and hotspot detection system.

Figure 6:Architecture of the proposed air quality prediction

Figure shows the overall architecture of the proposed system. The process begins with the environmental data, then it undergoes preprocessing and feature engineering. The processed data are employed to learn and compare machine-learning models. The selected Random Forest model is explained using SHAP, and the spatial hotspot analysis is conducted using K-Means clusters.

3.4 Project Plan and Milestones

The project was developed through a sequence of interconnected milestones.

Table 2:Project Plan and Milestones

Project phase	Planned activity	Evidence of completion / actual progress	Change or management action
Phase 1	Research planning and literature review	Research problem, objectives and literature base established	Literature review subsequently strengthened following feedback
Phase 2	Dataset collection	DEFRA UK-AIR and AURN geographical datasets obtained	Secondary geographical data retained specifically for spatial analysis

Phase 3	EDA	Pollutant distributions, temporal trends and monitoring-station activity analysed	NO ₂ selected as target based on data availability
Phase 4	Preprocessing	Timestamp conversion, numeric conversion, pivoting, cleaning and geographic integration completed	Records without required geographic coordinates excluded
Phase 5	Feature engineering	Temporal and historical features created	Later validation identified the need to reassess target-derived historical features
Phase 6	Initial ML modelling	Four regression models trained and compared	Additional validation experiments introduced after unusually high R ² was identified
Phase 7	Explainability	SHAP analysis completed	Interpretation revised to distinguish model contribution from causality
Phase 8	Spatial analysis	K-Means clustering and hotspot classification completed	DBSCAN was not retained because of computational cost
Phase 9	Visualisation	Folium hotspot map implemented	Basemap configuration refined during implementation
Phase 10	Flask development	Local prediction application implemented	Deployment model changed from original seven-feature model to corrected six-feature model
Phase 11	Validation	Additional feature-validation experiments and application scenarios completed	Corrected Random Forest used for final Flask prototype

Project management followed an iterative approach across the dissertation as opposed to a strict follow-through of the first workflow. The original workflow involved research planning and literature review, dataset acquisition, exploratory data analysis (EDA), preprocessing, feature engineering, machine learning modeling, explainability, spatial analysis, visualization, and Flask implementation. The technical aspect was progressing weekly from dataset exploration and EDA to preprocessing and feature engineering, to model development. Technical documentation supports the fact that the dataset used in the model was engineered with historical and temporal variables before the modeling stage.

Implementation phase required some changes as well. The spatial analysis was performed using DBSCAN algorithm until it became too computationally heavy with the amount of data available.

The first interface had seven features to predict which included rolling mean. The corrected implementation includes six features: Hour, Day, Month, Weekday, corrected NO₂ Lag 1, and corrected NO₂ Lag 2. It is consistent with the corrected random forest model.

3.5 Dataset Description

Table 3:Dataset Summary

Dataset	Source	Main information	Size/structure	Purpose
airQuality_2024_data.csv	DEFRA UK-AIR	Pollutant measurements, time, monitoring station and site information	1,048,575 records × 10 attributes	EDA, preprocessing, feature engineering and machine learning
AURN_2015_2023.csv	Kaggle / AURN	Monitoring station and geographical information	22 columns	Latitude/longitude extraction, spatial clustering and hotspot mapping

However, both datasets were retained due to their different uses. While the main dataset holds the information needed for NO₂ prediction, the secondary dataset holds geographical coordinates that are needed in spatial analysis.

<https://www.kaggle.com/datasets/airqualityanthony/uk-defra-aurn-air-quality-data-2015-2023>

Table 4: Primary Dataset Attributes

Attribute	Description
StartTime	Start time of pollutant measurement.
EndTime	End time of pollutant measurement
Status	Measurement validation/status information
Unit	Measurement unit
Value	Recorded pollutant concentration
PollutantName	Pollutant being measured
SiteName	Monitoring station
SiteType	Monitoring station type
Region	Geographic region
Country	Country associated with the station

These variables provide the data pertinent to time, environment, and monitoring stations' location for the subsequent analysis process. There are several pollutants present in the dataset. However, NO₂ was chosen as the pollutant of interest because it is the main pollutant analysed in the study.

3.6 Risk Management

Principles of risk management were introduced during the technical development stage in order to recognize any problems that could affect data quality, computational capability, model validity, and the development of applications. The main risks and their associated countermeasures are provided in Table .

Table 5:Project Risk Management

Risk	Potential impact	Mitigation / management action	Outcome
Large environmental dataset	Increased processing time and computational demand	Data cleaning and staged processing were used; spatial analysis was conducted using an appropriate	Analysis completed

		dataset/sample for clustering	
Missing or invalid pollutant values	Reduced modelling quality	Values were converted to numeric form, invalid values treated as missing and incomplete records removed	Clean NO ₂ dataset produced
Missing geographical coordinates	Prevented spatial integration	Records without required coordinates were excluded from geographic integration	Spatial dataset produced
DBSCAN computational cost	Delayed spatial analysis	K-Means was adopted as a computationally more practical alternative	Five-cluster analysis completed
Unusually high predictive performance	Risk of overestimating model validity	Additional experiments were conducted with and without historical target-derived features	Corrected performance established
Feature mismatch between model and application	Flask prediction failure	Model feature count and feature ordering were checked before deployment	Six-feature application successfully tested
Limited time for external user testing	No large-scale usability evaluation	Scenario-based technical testing was used instead	Four application scenarios tested
Local prototype deployment	Limited production readiness	Application was treated as a prototype and production deployment identified as future work	Appropriate limitation documented

3.7 Sampling Strategy

This research is based on secondary and observational environmental data without primary data from humans. Conventional participant sampling is not used for the machine learning dataset. The environmental observations are obtained from monitoring stations that are available and the periods for which measurements have been recorded. The spatial analysis uses monitoring stations for which geographical

coordinates are available. Geographical station records were not duplicated and observations not provided with the necessary coordinates were not included in the integration of datasets. However, due to the high computational cost of processing the entire dataset for spatial clustering, a representative sample of the integrated data was used for the K-Means clustering analysis.

3.8 Data Pre-processing

The data for the environment that was needed were raw and had to be transformed to be used in machine learning. The temporal information was obtained by converting the StartTime variable to datetime (Yang et al., 2025). Pollutant data were recoded to numeric data, and data that were not valid were recoded as missing data prior to discarding incomplete data. The original data were presented in long format with pollutant names and measurements as rows. A pivot transformation was therefore applied to develop a broader analytical structure. This facilitated separating NO₂ as the prediction target while retaining monitoring station and time information. The secondary geographical data was then trimmed to the needed station, latitude and longitude data (Koziel et al., 2024). Records with duplicate station numbers and missing coordinates were eliminated.

3.9 Feature Engineering

Feature engineering was used to represent both temporal behaviour and recent pollution history.

Four temporal features were extracted:

- Hour
- Day
- Month
- Weekday
- NO₂ Lag 1
- NO₂ Lag 2
- NO₂ Rolling Mean

3.9.1 Historical Feature Validation and Correction

After the model development process had been performed, an additional validation process was introduced because the Linear Regression model showed $R^2 = 1.0000$. It means that the historical target-derived features might have contained excessive information concerning the target under the conditions of the current experiment design. While the use of lag and rolling is not leakage by default as long as it does not include the use of any present or future target information in their creation, their construction and time splits have to be examined carefully.

Therefore, four different experiments have been run. In Experiment A, the same seven features have been used. Experiment B has discarded all lag and rolling variables and used only Hour, Day, Month, and Weekday.

Table 6: Feature and Purpose of Model Variables

Feature	Purpose
Hour	Represents within-day variation
Day	Represents day-related temporal variation
Month	Represents seasonal variation
Weekday	Represents weekday/weekend differences
NO ₂ _lag1	Provides the previous NO ₂ value
NO ₂ _lag2	Provides an additional historical value

3.10 Machine Learning Model Development

Four regression models were developed:

- Linear Regression
- Decision Tree Regressor
- Random Forest Regressor
- XGBoost

The use of various models was aimed at comparison of various model assumptions and justification of the final choice of the predictive model.

Linear Regression: Linear Regression was chosen as the base model due to its simplicity and interpretability. It serves as a benchmark against which other models could be compared (Yenkikar et al., 2025). Its main drawback lies in the possible non-linearity of the dependence between historical and temporal variables and NO₂.

Decision Tree: Decision Tree Regression was added as it can capture non-linear relationships by using recursive feature-based splits. It's somewhat easy to understand, but complex trees may overfit the training data.

Random Forest: Random Forest was added to make it more robust than a single decision tree. It merges several decision trees and can model more interesting relationships with decreased reliance on a single decision tree. The methodology refers to its processing capability of large datasets and complex relationships as relevant to air-quality prediction..

XGBoost: An advanced gradient-boosting method, XGBoost, was added. Unlike Random Forest, which takes advantage of independently generated trees, XGBoost builds the trees sequentially and later trees correct the errors of previous ones.

3.11 Train-Test Strategy

A time-aware data split was used for the modelling data. The data was not randomly shuffled as the prediction task also takes into account previously recorded NO₂ values. When following chronological order, there is less chance of modelling earlier observations using information from later observations. This is especially important in the presence of lag and/or rolling operations. The time-aware split is clearly defined as a way to keep data in temporal order and not to leak data. The implementation has an 80% training ratio and 20% testing ratio and the order remains the same. In order to further investigate the unusual initial performance of the model, the same time-dependent evaluation approach was used in additional validation tests.

3.12 Model Evaluation

The regression models were evaluated using:

Mean Absolute Error (MAE): The MAE is the mean absolute error between the predicted and observed NO₂. It offers an easy-to-use measure of prediction error.

RMS Error (RMSE): The RMSE is the square root of the mean squared prediction error. Errors of larger size have more impact on the RMSE and this makes it helpful to detect models that result in large prediction errors.

R²: R² represents the percentage accounted for by the model. Larger numbers represent higher model fit.

The usage of all three measures is a more comprehensive comparison of prediction quality than using any single quality measure.

3.13 SHAP Explainable AI

SHAP was chosen because predictive accuracy is not sufficient to explain the reason for a model to yield a specific NO₂ estimate. The approach assigns contribution values to features and can give both an overall sense of feature importance and reasons for individual predictions (Birinci et al., 2026; Houdou et al., 2024). The selected Random Forest model was applied to the selected SHAP model with its tree-based structure.

3.14 Spatial Clustering and Hotspot Detection

K-Means was used to analyse the spatial distribution of monitoring stations using latitude and longitude. Many factors were taken into account, such as the ability of DBSCAN to discover irregular cluster shapes and not needing to know the number of clusters beforehand. The methodology, however, reports that the available dataset required a significant amount of processing and memory usage for the DBSCAN, making it impractical for the project (Choi et al., 2024).

Therefore, K-Means was chosen as it offers a more efficient clustering method for the data available. However, K-Means uses distance calculations, so latitude and longitude were standardised before clustering. This is the current implementation with 5 clusters. The five-cluster configuration is a useful geographical grouping for monitoring locations. It does not imply that there are five pollution zones exactly in the UK defined scientifically (Yasin et al., 2025).

3.15 Folium Interactive Mapping

The spatial clustering results were presented using the Folium library and an interactive map. The spatial results are also not well represented by a numerical cluster label, so the interactive map is used for this information as well. The output is a `uk_no2_hotspot_map.html`, which was embedded into the Flask application. In this way, the existing spatial analysis was not repeated if the app was loaded again, avoiding the clustering process. This map shows the location of monitoring stations and the clusters to which they belong.

3.16 Flask Application Implementation

Flask was chosen as the web framework, as it allows the connection of an interactive user interface with a trained machine learning model in a lightweight structure. A Python-based backend is used, and the templates are HTML-based. The backend hosts the trained random forest model, processes the user input and outputs the NO₂ prediction, while the front end offers the user interface for the input of values and display of results.

The trained model was serialised as follows:

- `rf_model_corrected_model.pkl`

This avoids retraining the model every time the Flask application starts. The saved model may then be loaded and directly utilised for prediction, eliminating needless computation during application use.

The prediction interface has the same six features used during model development:

- Hour, Day, Month, Weekday, NO₂_lag1, and NO₂_lag2

The model's structure and ordering has to remain because the deployed model requires the same structure of input that was used when it was trained.

3.17 Ethical Considerations

Secondary environmental monitoring data, not personal or medical monitoring data, is used for the main analysis. The project does not require personal information in the machine learning part. For this project, no primary data from human participants are collected. The analysis is based on secondary environmental monitoring data, and the Flask prototype is tested technically to prediction and visualisation capabilities. Interpretation is also a critical function for which the responsibility must be exercised. The Flask prediction is not an environmental measurement, but a model estimate of NO₂ concentration. Similarly, the K-Means analysis does not establish a pollution hotspot per se, but rather a relative hotspot to identify which of the available monitoring clusters has a relative hotspot.

Chapter 4: Results and Discussion

4.1 Introduction

The results of the analysis of air quality data from the UK and the implementation of the environmental monitoring system are presented and discussed critically in this chapter. The results include exploratory data analysis, feature engineering, machine learning performance, hotspot identification, interactive visualisation and implementation in Flask application. The findings are then explained in terms of the research purpose and contrasted to the findings of other studies carried out in the field.

4.2 Dataset and Exploratory Data Analysis Findings

Exploratory data analysis (EDA) was conducted to get an overview of the primary features of the air-quality data before the predictive modelling. The analysis focused on pollutant distribution, pollutant frequency, monthly variation, differences between pollutant types and monitoring-station activity. This original data set has 1048,575 rows of data and 10 columns of data, including NO₂, PM10, O₃ and SO₂ measurements taken at monitoring stations.

4.2.1 Pollutant Frequency

The pollutant frequency analysis shows that NO₂ has the highest number of observations, followed by PM10 and O₃, while SO₂ has comparatively less number of observations.

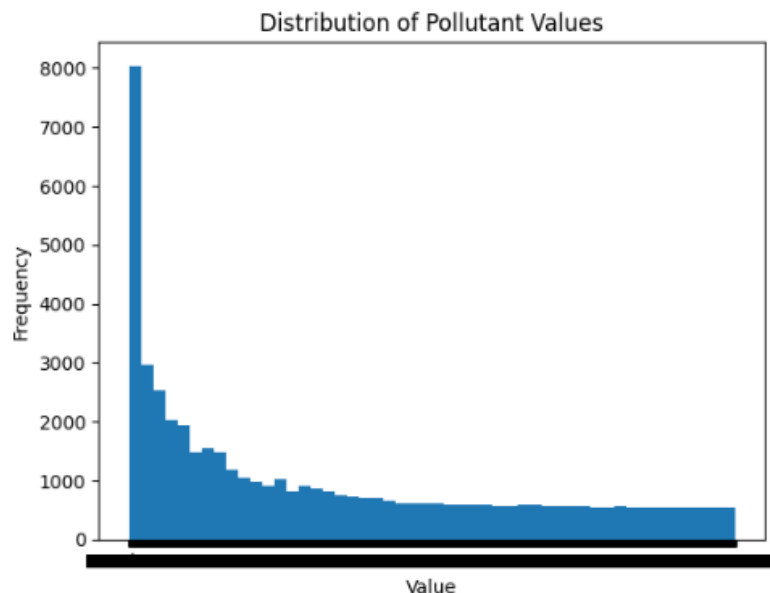


Figure 7: Frequency Distribution of Pollutants in the Air-Quality Dataset

The above figure shows the number of observations available for each pollutant is plotted. The distribution of observations for NO₂ is the largest and offers a good source of data for the prediction task. There are also significant observations of PM10 and O₃

and fewer observations for SO₂. The distribution shows that the pollutants are not equally distributed in the original data.

4.2.2 Distribution of Pollutant Values

The histogram examines the distribution of 50,000 sampled pollutant measurements from the dataset.

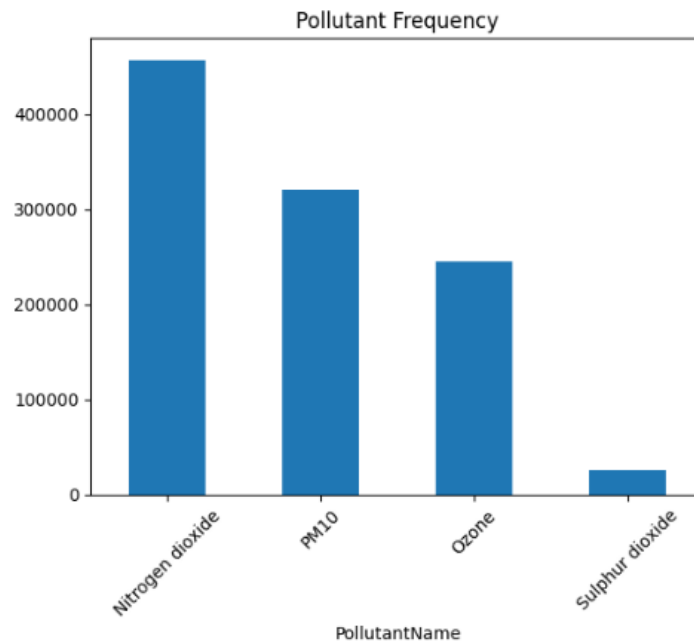


Figure 8: Distribution of Pollutant Concentration Values

The above figure shows the distribution of pollutant concentration values. This means that most observations have relatively low concentrations, and a smaller number of observations are carried towards the high concentrations.

4.2.3 Monthly Pollution Trend

Monthly aggregation was used to examine longer-term changes in pollutant concentration while reducing short-term fluctuations.

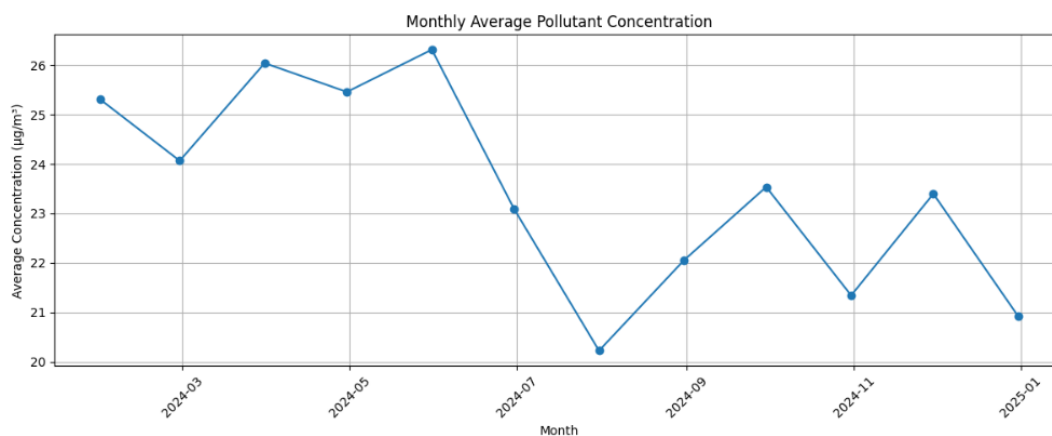


Figure 9: Monthly Average Pollutant Concentration

The above figure represents the changing pollution levels for each month separately. It clearly indicates that there is a temporal variation in pollutant concentration and not a constant one for the entire year. Therefore, using Month, Day, Weekday and Hour as temporal variables is justified.

4.2.4 Pollutant Distribution Comparison

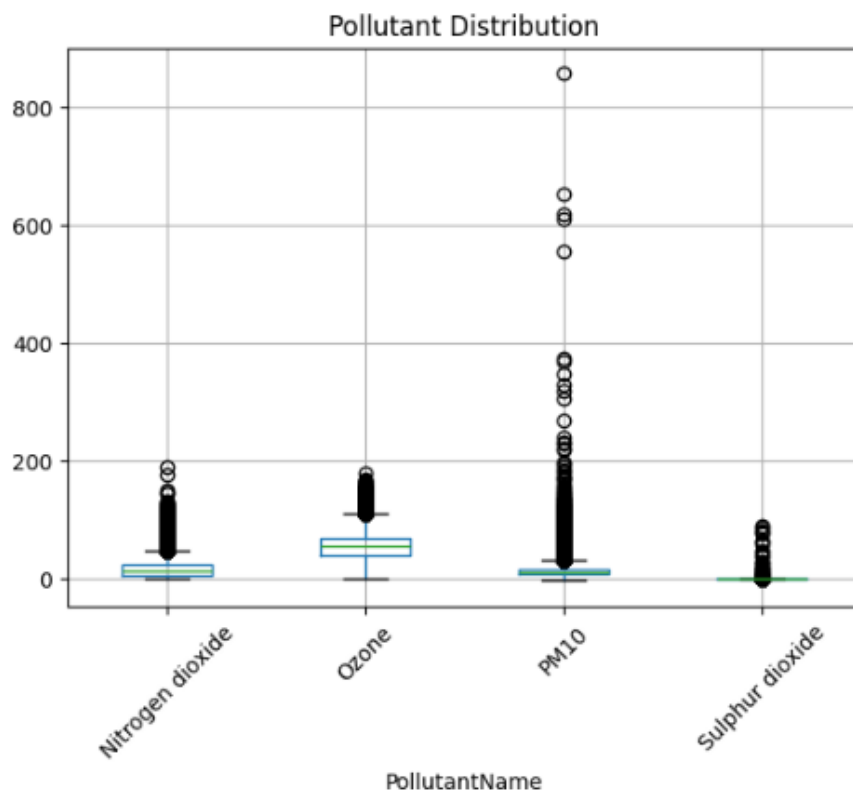


Figure 10: Comparative Distribution of Pollutant Concentrations by Pollutant Type

The figure represents a comparison of distributions of various pollutants through box plots. Differences in terms of medians, spread, and extremes exist among various pollutants. There are measurements which are found to be outside the main distribution ranges, suggesting that there might be some high or abnormal measurements.

4.2.5 Monitoring Station Activity

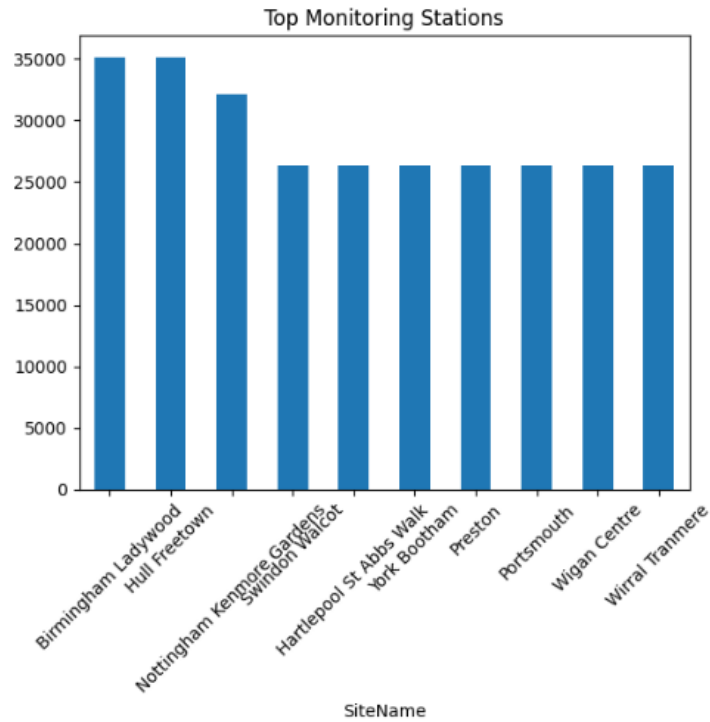


Figure 11: Top Monitoring Stations by Number of Recorded Observations

Monitoring sites with the largest number of observation records are shown in the above figure. It is found that the monitoring work at these sites differs in terms of their contribution to the number of records. This is crucial for understanding the data because the observations are not equally contributed by each monitoring site.

4.2.2 Data Pre-processing

The preprocessing stage transformed the `startTime` to a datetime type and the `Value` to a numeric type. All numeric values that were invalid were treated as missing values. Next, records where no pollutant measurements were valid were deleted from the analysis. The dataset was then pivoted from a long dataset to a wide one with `SiteName` and `StartTime` indexes and `PollutantName` as the variable column. This transformation allowed us to separate Nitrogen dioxide as a single variable. Incomplete records were deleted, and the appropriate columns after choosing NO_2 were renamed to `timestamp` and `NO2`.

```

df_model.info()

<class 'pandas.core.frame.DataFrame'>
Index: 437517 entries, 0 to 476623
Data columns (total 3 columns):
#   Column      Non-Null Count  Dtype  
---  -
0   timestamp    437517 non-null  datetime64[ns, UTC]
1   SiteName     437517 non-null  object  
2   NO2          437517 non-null  float64  
dtypes: datetime64[ns, UTC](1), float64(1), object(1)
memory usage: 13.4+ MB

```

Figure 12: Cleaned NO₂ Dataset Before Geographic Integration

The pre-processed dataset (NO₂) is displayed in the Figure. Timestamp values were converted into datetime format, pollutant values were converted to numeric form and incomplete records were removed. The resulting dataset includes 437,517 complete observations of NO₂, which is structured for feature engineering and predictive modelling.

4.2.3 Dataset Merging

A second AURN dataset was used to obtain geographical information for the monitoring stations. The site, latitude and longitude were extracted and renamed to SiteName to match the SiteName and duplicate stations were removed. Finally, the geographical information was joined with the cleaned NO₂ dataset using SiteName as the join key. The records that did not have latitude or longitude were then eliminated. This step was to merge the predictive dataset with the spatial data needed for K-Means clustering and hotspot mapping.

Table 7: Dataset Transformation Results

Processing stage	Records	Main result
Original dataset	1,048,575	10 raw attributes
After NO ₂ cleaning	437,517	Valid timestamp, site and NO ₂ values
Final integrated dataset	428,509	Geographic and prediction features available

Overall, the process of data preprocessing and merging turned the initial environmental measurements into a structured database that could be used for predicting NO₂ emissions and conducting spatial analysis. The results of the EDA stage showed that there was some variability of pollutants over time.

4.3 Feature Engineering Findings

The cleaned and geographically enhanced data was used to create variables for NO₂ prediction through feature engineering. Temporal information was incorporated to capture recurrent time patterns, while historical NO₂ variables represented recent pollutant behaviour (Al Yammahi & Aung, 2023). Initial feature engineering produced seven variables; however, subsequent validation identified a potential target-leakage issue with the original NO2_roll3 feature. The feature was therefore excluded from the final prediction model, and the historical lag variables were reconstructed using previous observations only.

Table 8: Features Used for NO₂ Prediction

Feature	Category	Purpose
Hour	Temporal	Represents variation within the day
Day	Temporal	Represents daily variation
Month	Temporal	Represents monthly/seasonal variation
Weekday	Temporal	Represents differences across days of the week
NO2_lag1	Historical	Previous NO ₂ observation
NO2_lag2	Historical	NO ₂ observation from the previous two steps

Temporal variables such as hour, day, month and weekday were derived from the timestamp to represent time-dependent patterns. Historical features were then constructed to capture the sequential behaviour of NO₂ measurements. The corrected lag variables use previous observations from the same monitoring station, ensuring that the features represent information available before the prediction point.

The original NO2_roll3 feature was excluded from the final model because its original construction incorporated the current NO₂ observation, creating potential target leakage. This was investigated through controlled feature-validation experiments, which showed that correcting the historical features produced more moderate and credible model performance.

Therefore, the final prediction model uses **six features**, combining calendar-based information with station-specific historical NO₂ observations. These features form the

input configuration used by the final Random Forest model and Flask prediction interface.

```
df_model.head()
df_model.info()

<class 'pandas.core.frame.DataFrame'>
Index: 428509 entries, 28 to 437516
Data columns (total 12 columns):
#   Column      Non-Null Count  Dtype
---  ---
0   timestamp   428509 non-null  datetime64[ns, UTC]
1   SiteName    428509 non-null  object
2   NO2         428509 non-null  float64
3   latitude    428509 non-null  float64
4   longitude   428509 non-null  float64
5   hour        428509 non-null  int32
6   day         428509 non-null  int32
7   month       428509 non-null  int32
8   weekday     428509 non-null  int32
9   NO2_lag1    428509 non-null  float64
10  NO2_lag2    428509 non-null  float64
11  NO2_rol13   428509 non-null  float64
dtypes: datetime64[ns, UTC](1), float64(6), int32(4), object(1)
memory usage: 36.0+ MB
```

Figure 13: Final Engineered Dataset Used for NO₂ Prediction

The figure above shows the final dataset after performing feature engineering on the data. Temporal features along with historical NO₂ features like NO2_lag1, NO2_lag2, were generated to make predictions. The features give the machine-learning models the history of the past behaviour of NO₂ along with the time-related data.

Therefore, the final set of features gave both types of data - calendar-based as well as the history of NO₂ to the modelling phase. The importance of this point is highlighted later while analysing the model and SHAP results.

4.4 Machine Learning Results

There were four regression models built to forecast NO₂ concentrations: Linear Regression, Decision Tree, Random Forest, and XGBoost. The performance of the models was measured using MAE, RMSE, and R². Based on the feature validation, the final set of models was constructed using NO₂ historical lags obtained only from the information available before the forecasting point.

At the first stage of model development, unexpectedly good results were achieved with the R² = 1.0000. The further analysis showed that there could be target leakage in the original feature NO2_rol13, which contained the current value of NO₂.

4.4.1 Model Performance

Table 9: Comparative Performance of the Corrected Regression Models

Model	MAE	RMSE	R ²
Linear Regression	4.0434	6.1986	0.8405
Decision Tree	6.1973	9.3013	0.6409
Random Forest	3.8748	5.9533	0.8529
XGBoost	3.9731	6.0510	0.8480

From the analysis, it is evident that the predictive power of the four models used is very high. In particular, all the three models-Decision Tree, Random Forest, and XGBoost-have R² greater than 0.98, which means that there is a huge proportion of variation in NO₂ concentration explained by the models.

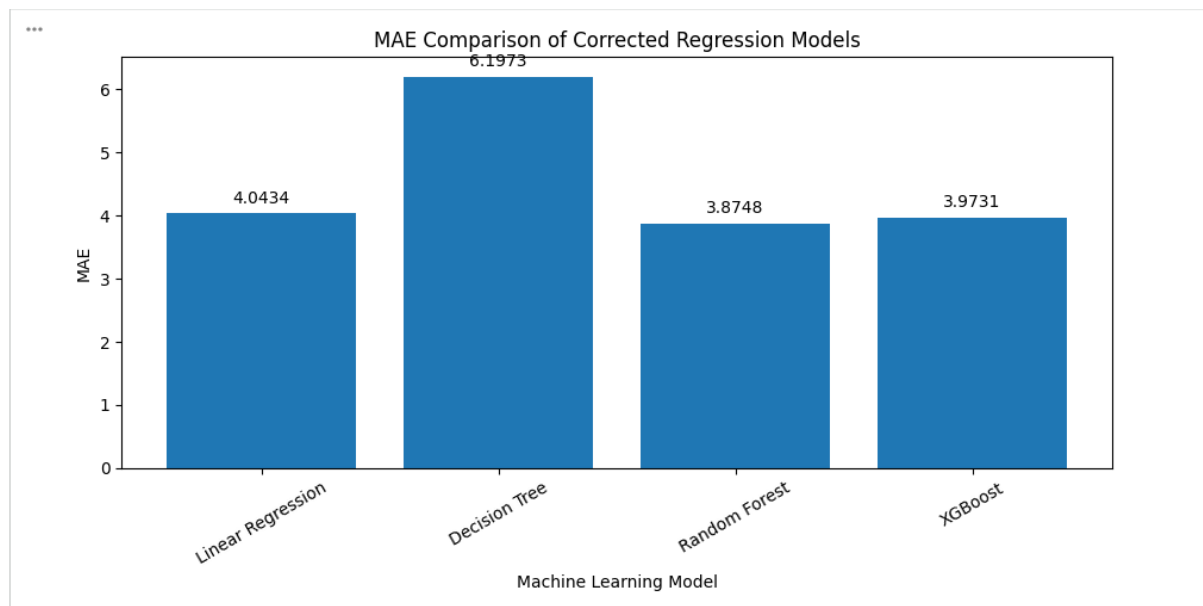


Figure 14: Comparative Performance of the Four NO₂ Regression Models

Figure depicts the results of the four models on the parameters MAE, RMSE, and R². Random Forest had the minimum error in prediction, while XGBoost and Linear Regression showed fairly good results. The Decision Tree had the highest value of MAE and RMSE and the minimum R². Initially, the results obtained were anomalous, with an R² score of 1.0000 for Linear Regression. It was later found that there might be a case of target leakage due to the NO₂_roll3 parameter, which included the current NO₂ reading.

4.4.2 Linear Regression Results

Linear Regression with corrections resulted in an MAE value of 4.0434, an RMSE value of 6.1986, and R² value of 0.8405. This model acted as a benchmark to evaluate the tree-based models.

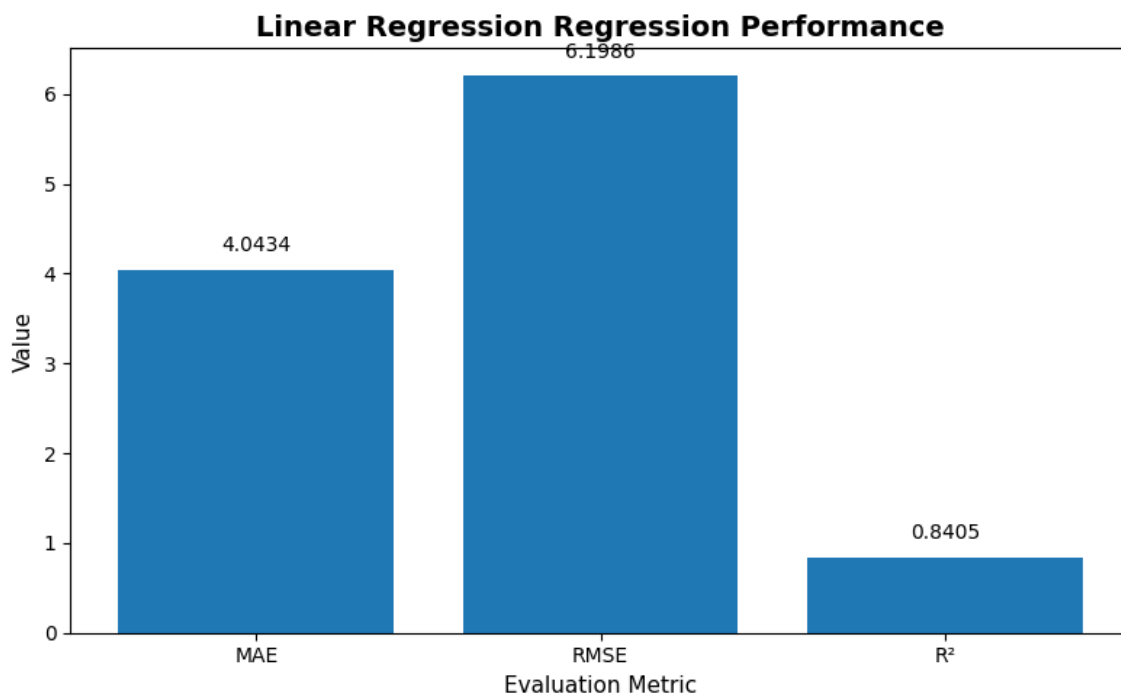


Figure 15: Linear Regression Performance

As shown, Linear Regression is capable of explaining a large portion of variability in NO₂ concentration using corrected temporal and historical features. The R² score of 0.8405 implies that about 84% of variability in NO₂ concentrations in the test set can be explained by the model.

Still, Linear Regression does not have enough capacity to capture non-linear relationships, therefore it provides a foundation to compare it with Decision Tree, Random Forest, and XGBoost models.

It should be mentioned that the first result of Linear Regression (R² = 1.0000) cannot be considered as the final performance of the model, since feature generation contained the NO₂_roll3 feature that is related to the target and causes target leakage.

4.4.3 Decision Tree Results

With regard to the Decision Tree, it gave the following measures: MAE = 6.1973, RMSE = 9.3013, and $R^2 = 0.6409$. This indicates that the relationship between the historical and temporal NO₂ attributes and the target concentration was well captured by the model, though it performed less efficiently compared to the other models analyzed.

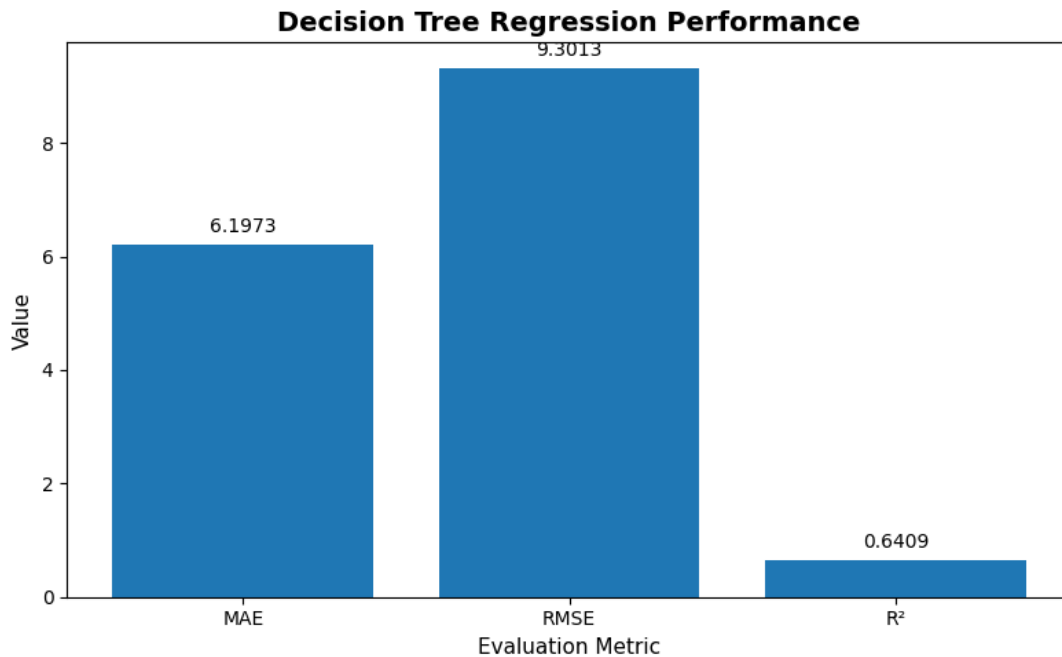


Figure 16: Decision Tree Regression Performance

Decision Tree had the largest mean absolute error (MAE) and root mean squared error (RMSE) and the smallest coefficient of determination (R^2) among all the corrected models. MAE means that the mean prediction error is large, and RMSE means that the prediction errors are large and have an even more significant effect on the model performance. The value of R^2 equals 0.6409, which implies that about 64% of NO₂ test concentrations variance is explained by the model.

4.4.4 Random Forest Results

The first experiment using the Random Forest algorithm had an MAE of 1.2301, RMSE of 1.8318, and R^2 of 0.9861, all of which suggest that the predictions have performed strongly under the original set of features. However, the unexpectedly high performance levels exhibited by the models during the early experiments have made it necessary to conduct a more thorough investigation of the process used in creating the historical NO₂ features. This is because the original rolling feature includes the

current NO₂ value, thus raising the risk of target leakage.

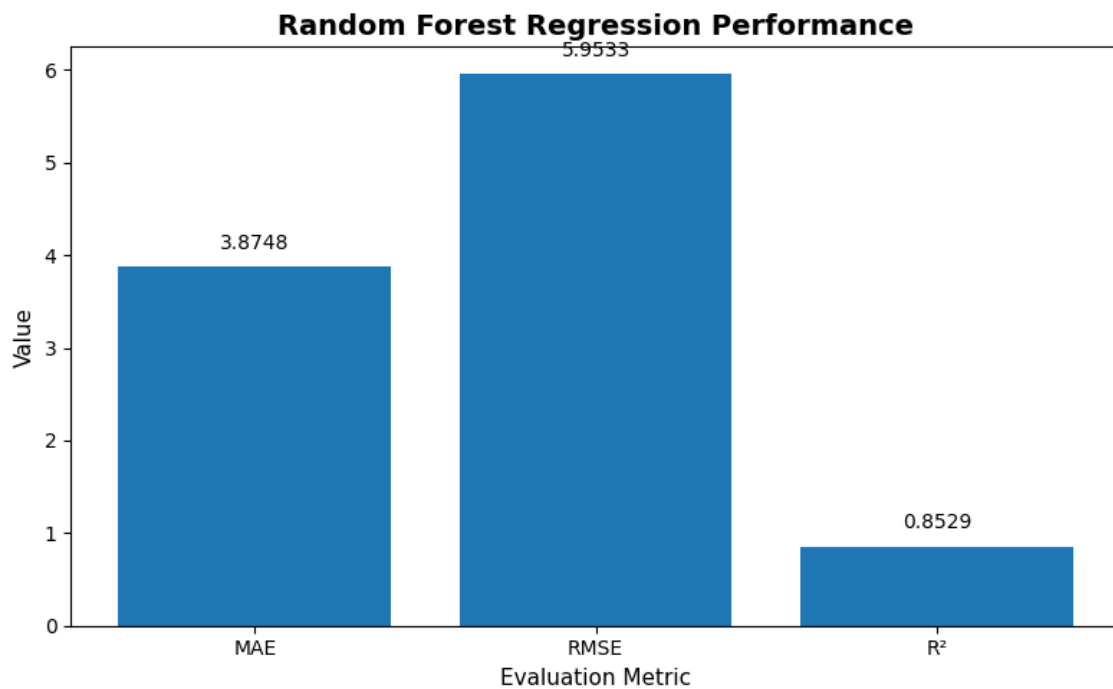


Figure 17: Random Forest Regression Performance

Having completed the supplementary validation procedures, the historical NO₂ features were created using solely the data known prior to the prediction time. The corrected Random Forest algorithm utilizing Hour, Day, Month, Weekday, corrected NO₂ Lag 1, and corrected NO₂ Lag 2 resulted in a MAE of 3.8748, an RMSE of 5.9533, and R² of 0.8529. Even though these results do not compare to the initial ones, they provide a more credible estimation by restricting feature creation in order to eliminate the leakage to the target observation.

Among all four models, the corrected Random Forest provided the best performance of all models. It had the smallest MAE (3.8748), RMSE (5.9533), and the largest R² (0.8529). To illustrate, the Linear Regression, Decision Tree, and XGBoost had R² equal to 0.8405, 0.6409, and 0.8480 correspondingly.

4.4.5 XGBoost Results

The XGBoost model, after corrections, gave MAE = 3.9731, RMSE = 6.0510, and R² = 0.8480. This shows that XGBoost performed very well in terms of prediction and was still in the race with the other models after corrections.

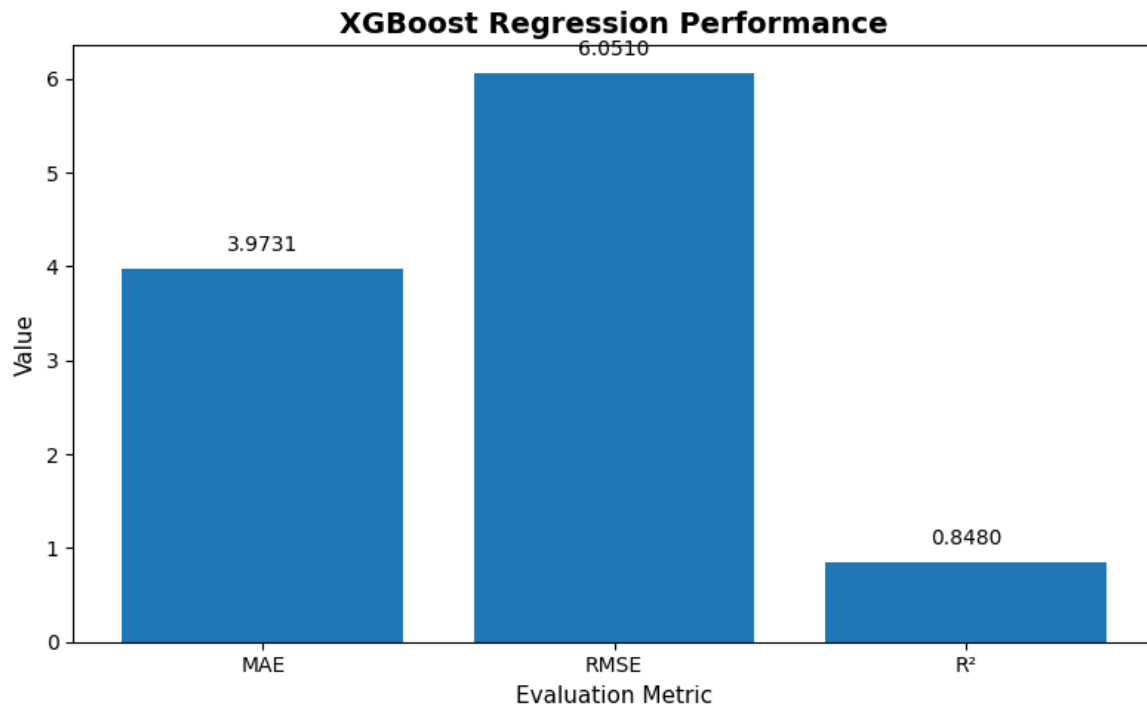


Figure 18: XGBoost Regression Performance

The figure illustrates the results of XGBoost's predictions using three performance measures. The predictions of the model have been relatively accurate, and considering the value of the coefficient of determination R^2 , which equals 0.8480, the model can explain quite a significant amount of variability in the concentrations of NO_2 in the test set.

XGBoost uses sequential decision trees, where the trees are built to compensate for the mistakes made by the preceding trees. This way, the model is able to identify complicated nonlinear dependencies between various historical and temporal characteristics of NO_2 .

4.4.6 Overall Result

The improved results show that Random Forest had the best predictive power among all the models, with MAE = 3.8748, RMSE = 5.9533, and R^2 = 0.8529. The XGBoost and Linear Regression models had acceptable results, while the Decision Tree showed the largest prediction errors and the smallest R^2 value.

The results obtained at the first stage were very close to perfect, which made us look for a possible target leakage issue related to the original $\text{NO}_2_{\text{roll3}}$ variable. After removing this variable and re-constructing the historical lag variables based only on past values, we got the improved results.

Given the corrected evaluation of model performance, we chose Random Forest as the final predictive model for SHAP interpretation and Flask integration. This choice was mostly motivated by the performance of the model in terms of MAE, RMSE, and R^2 metrics, as well as explainability and applicability issues.

4.5 Additional Model Validation Experiments

This modelling phase showed outstanding predictive results, as Linear Regression achieved R^2 of 1.0000, while the tree-based algorithms achieved R^2 higher than 0.98. Although the performance of models appeared to be extremely accurate, the large value of Linear Regression result made researchers doubt whether the inclusion of target-derived historical features played any role. In order to determine whether such anomalous prediction was related to those historical features, an additional validation phase was introduced. In total, four controlled experiments were carried out, including Experiment A with the original set of features; Experiment B without lag and rolling variables; Experiment C with corrected historical features; and Experiment D only with corrected lag features.

4.5.1 Original Feature Configuration

Experiment A reproduced the original modelling configuration used in the initial analysis. The feature set consisted of:

- Hour
- Day
- Month
- Weekday
- NO2_lag1
- NO2_lag2

The four regression models were evaluated using MAE, RMSE and R^2 . The results were broadly consistent with the original modelling stage.

Table 10: Configuration Regression Models

Model	MAE	RMSE	R^2
Linear Regression	0.0000	0.0000	1.0000
Decision Tree	0.8733	1.5863	0.9896
Random Forest	1.2500	1.8617	0.9856
XGBoost	0.8373	1.7897	0.9867

The results of the Linear Regression had a clear outlier nature, with zero value for both MAE and RMSE, as well as with R^2 being 1.0000. Even though this result would at first appear to indicate very good prediction capability, such results are not automatically considered to be proof of perfect generalization. Therefore, the feature construction process needed further examination.

4.5.2 Model without Historical NO₂ Features

Experiment B was designed to determine how much predictive information was provided by the historical NO₂ variables. The lag and rolling variables were removed, leaving only:

- Hour
- Day
- Month
- Weekday

The results changed substantially.

Table 11: Futuristic Models

Model	MAE	RMSE	R ²
Linear Regression	12.1358	17.2855	-0.2402
Decision Tree	11.4259	15.4248	0.0125
Random Forest	11.2161	15.4608	0.0078
XGBoost	10.9820	14.8476	0.0850

Given the drop in performance, it is evident that the temporal variables did not give any predictive power about the NO₂ measurement. In this case, the R² for the Random Forest fell from 0.9856 in Experiment A to 0.0078 in Experiment B, while that of XGBoost was 0.0850. It is important to note that the negative R² generated by Linear Regression suggests that its predictions were less accurate than using the mean target value as a baseline.

4.5.3 Corrected Feature Configuration

After the diagnosis was performed, historical information was recreated such that it would not use information gained after the time point of prediction but only the information known beforehand.

The corrected features were created as follows:

- NO₂_lag1_corrected: previous NO₂ observation
- NO₂_lag2_corrected: NO₂ observation from two previous observations

The corrected historical features were evaluated in Experiment C. The results were:

Model	MAE	RMSE	R ²
Linear Regression	4.0409	6.1963	0.8407
Decision Tree	6.0711	9.1452	0.6529

Random Forest	3.8702	5.9468	0.8532
XGBoost	3.9503	6.0133	0.8499

These results show a significant improvement compared to Experiment B, but they are still considerably lower compared to the original results obtained in Experiment A.

For Random Forest, the result is $R^2 = 0.8532$ with MAE = 3.8702 and RMSE = 5.9468. For XGBoost, the result is almost identical, where $R^2 = 0.8499$. The fact that there is such similarity between the two methods indicates that the corrected historical features still contain high levels of predictability information.

Another experiment, Experiment D, was performed using only the corrected lag features without the corrected rolling feature. The results were:

Model	MAE	RMSE	R^2
Linear Regression	4.0434	6.1986	0.8405
Decision Tree	6.1973	9.3013	0.6409
Random Forest	3.8748	5.9533	0.8529
XGBoost	3.9731	6.0510	0.8480

The output for the random forest model for Experiment D was virtually the same as for Experiment C, where the R^2 moved from 0.8532 to 0.8529, while the RMSE moved from 5.9468 to 5.9533. It is apparent that the new rolled up attribute added only little extra information in comparison to the lagged ones.

Consequently, the corrected Random Forest using the six features below was selected for the final Flask prototype:

1. Hour
2. Day
3. Month
4. Weekday
5. Corrected NO₂ Lag 1
6. Corrected NO₂ Lag 2

The rolling-mean variable was therefore excluded from the final deployed model.

4.5.4 Comparative Validation Results

The four experiments demonstrate how strongly model performance depended on the construction of the historical NO₂ variables.

Table 12: Validation results

Experiment	Feature configuration	Model	MAE	RMSE	R ²
A	Original features	Linear Regression	0.0000	0.0000	1.0000
A	Original features	Random Forest	1.2500	1.8617	0.9856
B	Without lag/rolling	Linear Regression	12.1358	17.2855	- 0.2402
B	Without lag/rolling	Random Forest	11.2161	15.4608	0.0078
B	Without lag/rolling	XGBoost	10.9820	14.8476	0.0850
C	Corrected historical features	Random Forest	3.8702	5.9468	0.8532
C	Corrected historical features	XGBoost	3.9503	6.0133	0.8499
D	Corrected lag features only	Random Forest	3.8748	5.9533	0.8529
D	Corrected lag features only	XGBoost	3.9731	6.0510	0.8480

The results obtained from using the first seven-feature combination were quite outstanding, including Linear Regression's optimal performance. Secondly, deleting historical information about NO₂ almost eliminated the predictive performance of the tree models, implying that the current observations about NO₂ formed crucial predictors. Lastly, correcting historical features resulted in significant improvement in predictive performance, where Random Forests managed to obtain an R² of around 0.853.

The similarities between Experiments C and D were striking. By not including the corrected rolling feature, there was no considerable drop in the performance of Random Forests since its R² dropped from 0.8532 to 0.8529.

4.5.5 Discussion of Potential Data Leakage

The additional experiments show an important problem in the first implementation of the NO2_roll3 feature. Namely, the feature was implemented as follows:

```
df_model['NO2_roll3'] = df_model['NO2'].rolling(window=3).mean()
```

The rolling function was performed on the target NO2 series without any preceding shifting, therefore the rolling window included the current NO2 observation. As the current NO2 observation is also the target to be predicted, the feature included information about the target observation. It is a leakage by construction, not just a potential leakage.

This is the possible explanation for the unusual good performance of Experiment A, especially the result $R^2 = 1.0000$ of the Linear Regression model. This problem is not in the fact of using rolling features in general; a rolling feature is a correct choice in a forecasting framework if it is calculated using only past observations.

The new approach fixes the problem by calculating historical features based on past observations, making sure the lag variables include only past information. The new rolling feature was also calculated based on past observations without including the current target. Thus, $R^2 = 1.0000$ cannot be taken as proof of the ability to make perfect predictions in the real world; it must be understood as a result that helps identify a flaw in the feature construction process. The additional experiments give a much more reliable estimate of the performance of the model.

Critical Model Comparison

The models have been evaluated using MAE, RMSE, and R^2 . The lower MAE and RMSE values indicate reduced prediction error, while higher R^2 indicates increased explanation power. These measures are applicable to the target variable that is NO₂ concentration, which is a continuous variable.

Table 13: Critical Comparison of Corrected Model Performance

Model	MAE	RMSE	R^2	Critical interpretation
Linear Regression	4.0434	6.1986	0.8405	Strong baseline performance
Decision Tree	6.1973	9.3013	0.6409	Weakest corrected performance
Random Forest	3.8748	5.9533	0.8529	Strongest overall performance; selected for SHAP and deployment
XGBoost	3.9731	6.0510	0.8480	Competitive performance

The first set of tests gave an extremely good performance result, such as R^2 of 1.0000 for Linear Regression. However, the need for a deeper analysis of historic features became inevitable. Initially, there was a feature called NO2_roll3 that included present

NO₂ and therefore created a target leakage. Therefore, the feature was deleted, and the historic lag features were redefined based on previous observations.

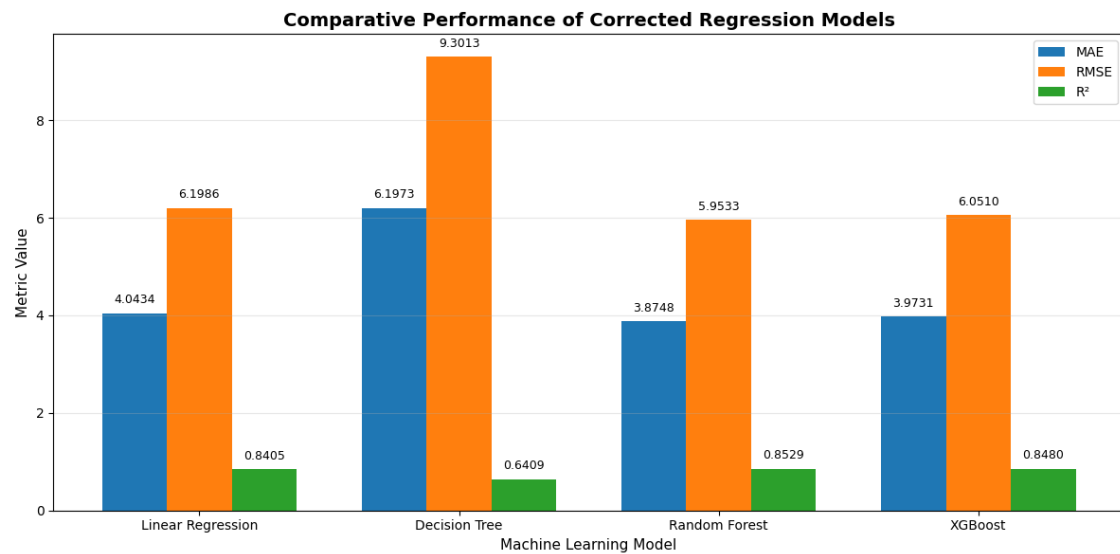


Figure 19: Comparative Performance of the Four Regression Models

All four regression models are compared based on MAE, RMSE and R². In terms of numerical results, Linear Regression gave the best result, and XGBoost gave the lowest MAE for non-linear models. Random Forest was chosen for practical implementation due to its support for the project's explainability needs via SHAP and the Flask integration requirements.

4.6 Actual vs Predicted Results

To visually evaluate the degree of accuracy of the Random Forest model in replicating NO₂ concentration observed, Actual versus predicted analysis was performed. The actual values of NO₂ from the test dataset are compared with the values predicted by the trained Random Forest model. The technical implementation is a plot of the initial 200 test observations to facilitate visual interpretation of the comparison.

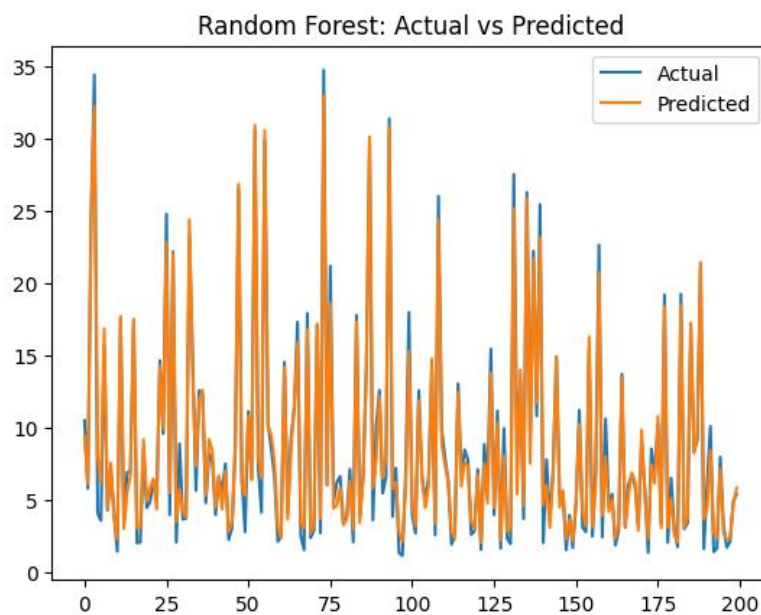


Figure 20: Random Forest Actual and Predicted NO₂ Concentrations

The figure shows two series is the actual NO₂ measurements and the Random Forest predictions. If the predicted line is close to the actual line, the model is doing a good job reproducing the observed concentration. The differences between the two lines show the prediction errors: the larger the difference, the more difficult it is for the model to reconcile the measured concentration.

4.7 SHAP Results and Interpretation

To understand the contribution of each of the input variables to the NO₂ concentration predicted by the model, SHAP (SHapley Additive exPlanations) was used to interpret the Random Forest model. This analysis was significant because the prediction output doesn't describe why a specific NO₂ value was generated. SHAP offers a feature-level explanation of how the model is working by displaying the relative importance of each predictor (Houdou et al., 2024).

4.7.1 SHAP Summary Results

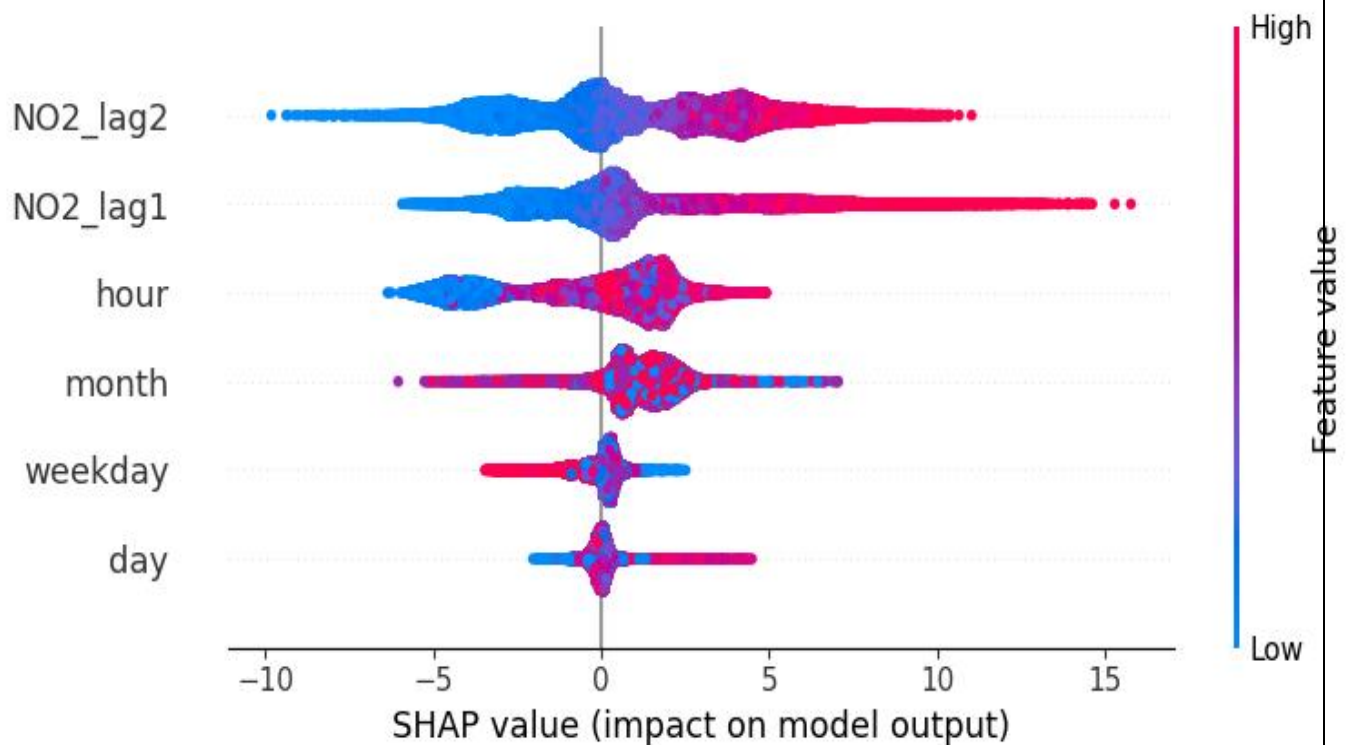


Figure 21: SHAP Summary Plot Showing Feature Contributions to Random Forest

The SHAP summary plot in Figure 21 illustrates the contribution of the six features used by the final Random Forest model. The features are ranked according to their overall influence on the model predictions. NO2_lag2 and NO2_lag1 have the greatest influence, indicating that recent historical NO₂ concentrations are the strongest predictors of NO₂ concentrations. Among the temporal variables, hour has a comparatively greater influence, followed by month, weekday and day. The distribution of SHAP values also shows that individual features can either increase or decrease the predicted NO₂ concentration depending on their values.

Importantly, this SHAP analysis corresponds to the corrected six-feature Random Forest model after the original NO2_roll3 feature was removed because of target leakage. Therefore, NO2_roll3 is not included in the final SHAP interpretation. The dominance of NO2_lag2 and NO2_lag1 suggests that recent historical NO₂ measurements have a greater influence on the model predictions than the calendar-based variables. However, SHAP values describe how features contribute to the model's predictions and should not be interpreted as evidence of causal relationships.

4.7.2 Interpretation of SHAP Values

The horizontal SHAP axis represents the contribution of individual feature values to the model prediction. Positive SHAP values indicate an increase in the predicted NO₂

concentration, whereas negative values indicate a decrease relative to the model's baseline prediction. The colour scale represents the magnitude of each feature value, with red indicating higher values and blue indicating lower values. The plot therefore provides information about both feature importance and the direction of feature contributions. However, these results should be interpreted as explanations of the model's behaviour rather than as evidence of causal relationships (Jiménez-Navarro et al., 2024). The strong SHAP contributions of NO₂_lag1 and NO₂_lag2 indicate that the Random Forest relies heavily on recent historical NO₂ measurements when generating predictions, but this does not establish that these variables causally determine subsequent NO₂ concentrations. Overall, the SHAP analysis improves the transparency and interpretability of the final six-feature model and shows that its predictions are influenced more strongly by recent NO₂ history than by the temporal variables (Birinci et al., 2026).

4.8 K-Means Hotspot Results

The K-Means clustering method was applied to the monitoring sites based on the standardized latitudes and longitudes to group the geographical regions and then compare their average NO₂ concentration levels. There are five clusters. The objective was not prediction of pollution levels but finding relative differences between the clusters.

Table 14: K-Means Hotspot Analysis

Analysis	Result
Clustering method	K-Means
Spatial variables	Latitude and Longitude
Number of clusters	5
Hotspot cluster	Cluster 3
Average NO ₂ concentration	≈18.71 µg/m ³

```

Mean NO2 concentration by cluster:
Cluster
3      18.705696
4      17.505082
1      16.655271
0      14.715663
2      14.301977
Name: NO2, dtype: float64

Hotspot Cluster: 3
Mean NO2: 18.71 µg/m³

```

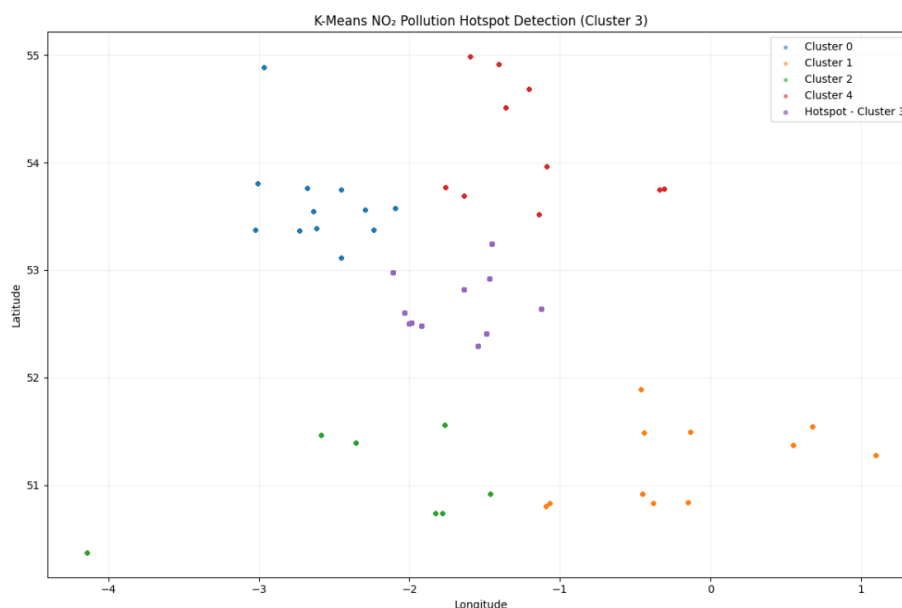


Figure 22: K-Means Pollution Hotspot Detection

The geographical clusters are shown in Fig. 4.15. Cluster 3 has been labeled as the relative hotspot because it contains the highest mean value of NO_2 concentration (around $18.71 \mu\text{g}/\text{m}^3$).

4.9 Folium Spatial Findings

The K-Means analysis was plotted using Folium to create an interactive geospatial mapping of the monitoring sites and the clusters created. The map created is stored as `uk_no2_hotspot_map.html` and added to the Flask application using an `iframe`. The map gives geographic context to the K-Means output and adds value to understanding the clusters compared to static outputs. Cluster 3 represents the cluster that has the highest average NO_2 concentration. But the map only shows the monitoring sites and the cluster each belongs to; it does not show the NO_2 concentration estimates of areas where there are no monitoring sites.

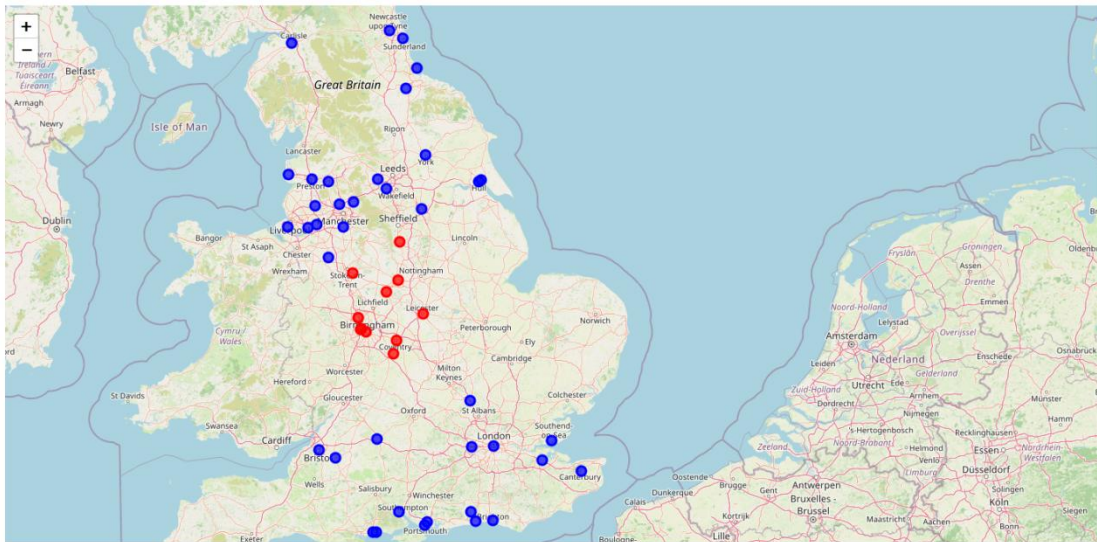


Figure 23: Interactive Folium Map of NO₂ Hotspot and Monitoring Locations

The above figure shows the interactive Folium map of the monitoring locations and K-Means clusters. The map displays a geographical overview of the spatial analysis and facilitates interpretation of the cluster distribution. It is not an estimate of pollution concentrations in areas without monitoring stations, but rather represents available monitoring stations. A map can be used to view the monitoring locations and differentiate the identified hotspot from the other monitoring clusters.

4.10 Flask Prediction Application

In the Flask application, there is an operational interface that can be used to deploy the Random Forest model without having to train it again. The corrected model is loaded from `rf_model_corrected.pkl` at the time when the application is launched. The prediction form uses the same six variables as the corrected model: Hour, Day, Month, Weekday, NO₂ Lag 1, and NO₂ Lag 2.

Figure 24: Environmental NO₂ Prediction Interface

This application shows the implementation of the refined machine learning algorithm in practice within a browser interface. It was evaluated using scenario testing in terms of normal scenarios, high levels of NO₂ in recent years, low levels of NO₂ in recent years, and wrong data scenarios. This is an application developed locally but not deployed to production.

4.11 Testing and Technical Refinement

For evaluating the efficiency of the Flask prediction app, scenario testing was carried out using various inputs, both valid and invalid. The testing was aimed at checking the efficiency of the app in making NO₂ predictions and handling invalid user inputs.

Table 15: Technical Refinements

Scenario	Test condition	Prediction / Response	Result
1. Normal conditions	Hour 14, Day 15, Month 6, Weekday 3, Lag 1 = 18.5, Lag 2 = 17.8	19.69 µg/m ³	Pass
2. High recent NO ₂	Hour 16, Day 15, Month 6, Weekday 3, Lag 1 = 60, Lag 2 = 55	56.44 µg/m ³	Pass
3. Low recent NO ₂	Hour 16, Day 15, Month 6, Weekday 3, Lag 1 = 6, Lag 2 = 5	6.81 µg/m ³	Pass
4. Invalid input	Hour = 25	Input rejected	Pass

From the findings, it is clear that the Flask application generated the predictions under different NO₂ scenarios and also refused an invalid hour input. The test shows how the prototype functions without any external evaluation from users.

The screenshot shows the 'EnviroPredict' application interface. On the left, the 'Prediction Inputs' form contains the following values: Hour (22), Day (15), Month (6), Weekday (3), NO₂ Lag 1 (60), and NO₂ Lag 2 (50). On the right, the 'Prediction Generated' box displays an 'Estimated NO₂ Concentration' of 49.26. Below this value, it states 'NO₂ concentration' and provides a note: 'The value above represents the estimated nitrogen dioxide concentration generated by the trained Random Forest regression model.'

Figure 25: Flask application with high NO₂ input

The screenshot shows the 'EnviroPredict' application interface with a 'Predict NO₂ Concentration' button at the bottom of the input form. The 'Prediction Inputs' form contains the following values: Hour (22), Day (15), Month (6), Weekday (3), NO₂ Lag 1 (4), and NO₂ Lag 2 (3). On the right, the 'Prediction Generated' box displays an 'Estimated NO₂ Concentration' of 3.97. Below this value, it states 'NO₂ concentration' and provides a note: 'The value above represents the estimated nitrogen dioxide concentration generated by the trained Random Forest regression model.'

Figure 26: Flask application with low NO₂ input

The screenshot displays the 'EnviroPredict' web application interface. On the left, the 'Prediction Inputs' section contains several input fields: 'Hour' (set to 24), 'Day' (set to 15), 'Month' (set to 6), 'Weekday' (set to 3), and 'NO₂ Lag 1'. A red error message 'Value must be less than or equal to 23' is shown next to the 'Hour' field. On the right, the 'Prediction Generated' section shows the 'Estimated NO₂ Concentration' as 3.97, with a note that this value represents the estimated nitrogen dioxide concentration generated by the trained Random Forest regression model.

Figure 27: Invalid input validation

4.12 Results Compared and Critiqued to Literature

The comparison of Linear Regression, Decision Tree, Random Forest, and XGBoost was conducted with MAE, RMSE, and R^2 as the performance metrics. Following the feature validation, Random Forest proved to be the best-performing among all algorithms, with the results being MAE = 3.8748, RMSE = 5.9533, and R^2 = 0.8529. The R^2 scores of Linear Regression, XGBoost, and Decision Tree were 0.8405, 0.8480, and 0.6409, correspondingly.

A preliminary result of R^2 equal to 1.0000 obtained by means of Linear Regression prompted the further investigation as the feature containing rolling NO₂ values contained the target value itself and there might have been the case of target leakage. Further experiments demonstrated the necessity of historical features and resulted in the correction of the features' configuration. Thus, the final results can be taken as credible and the feature construction importance is shown.

In turn, the employment of SHAP corresponds to the findings of other air-quality research. According to Houdou et al. (2024), SHAP was found to be one of the most popular tools for explanation and, thus, can be used for the purpose of improving the transparency of environmental prediction.

Chapter 5: Conclusions and Recommendations

5.1 Conclusion

This paper introduces an overall environmental monitoring system using UK DEFRA UK-AIR data, machine learning, SHAP explainability, spatial clustering, and the Flask application. Four models were evaluated based on the metrics of mean absolute error (MAE), root mean squared error (RMSE), and R^2 . Further experiments were conducted concerning the exceptionally high value of the original R^2 equals 1.0, as well as the leakage in the rolling NO_2 feature. Thus, the feature NO2_roll3 was dropped and the history features were fixed. The results of the corrected Random Forest model are $\text{MAE} = 3.8748$, $\text{RMSE} = 5.9533$, and $R^2 = 0.8529$, which became the final model for the application. The SHAP method was used to explain the model and calculate the importance of the input features with a special focus on the historical NO_2 features. The k-means algorithm allowed for clustering the monitoring stations into five clusters where Cluster 3 had the highest average NO_2 concentration - $18.71 \mu\text{g}/\text{m}^3$.

5.2 Contribution

This project is a comprehensive solution for the prediction of UK NO_2 that uses machine learning, explainability by SHAP values, spatial clustering, and a prediction model built with Flask technology. One of the major contributions of this research is additional validation of the historical features of NO_2 due to an unexpectedly high value of $R^2 = 1.0000$. It was found that there could be the case of target leakage in the original rolling feature, leading to the deletion of the original rolling feature and using the new lag features. The final Random Forest model produced the following results: $\text{MAE} = 3.8748$, $\text{RMSE} = 5.9533$, and $R^2 = 0.8529$.

5.3 Limitations

A number of limitations exist in the study. Firstly, most of the analysis depends on the DEFRA UK-AIR dataset and can thus be questioned in terms of applicability for other locations or time periods within the UK. Validation using an external testing source was not carried out. Flask was tested with four technical scenarios but not with real external users and thus is only a local prototype at present. K-Means clustering results in relatively grouped geographic locations as opposed to pollution hot spots that have been statistically confirmed. The final model also requires prior information about NO_2 observations.

5.4 Future Scope

Future research can focus on external validation via independent monitoring sites, longer temporal tests, and additional inclusion of more meteorological features.

Another area of future research can include time series models and spatiotemporal models that are different from what was used. Hotspot analysis using statistically valid approaches can be conducted to expand the spatial analysis performed.

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Appendix

Colab code link

<https://drive.google.com/file/d/12gMc1V1Dot-6X1szau3bMjevSeGKy2ob/view?usp=sharing>

Flask app link

<https://drive.google.com/file/d/1QQqUOLj3QNUudt8RnWHghWrMTJLxtV8w/view?usp=sharing>

Dataset link-AirQuality_2024

<https://drive.google.com/file/d/1agZ9Sp7pQtDEkXQqfpvAM6XFUayHDcLJ/view?usp=sharing>

UK DEFRA AURN Air Quality Data 2015-2023

<https://www.kaggle.com/datasets/airqualityanthony/uk-defra-aurn-air-quality-data-2015-2023>

readme file

<https://github.com/Akashj77/-Interactive-Mapping-Application-for-Environmental-Monitoring-and-Forecasting/tree/main>.

Appendix - B

